

Problem 1 (Statistical distance).

Given two distributions $(\mathcal{X}, \mathcal{Y})$ over a finite set Ω , their statistical distance (also known as variational or L_1 distance) is defined as

$$\Delta(\mathcal{X}, \mathcal{Y}) \triangleq \frac{1}{2} \sum_{\omega \in \Omega} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)|.$$

Where $\mathcal{X}(\omega) \triangleq \Pr[x \leftarrow \mathcal{X} : x = \omega]$ is the probability of getting output ω when sampling from $\mathcal{X}$.

(i) Show that the following is an equivalent definition:

$$\Delta(\mathcal{X}, \mathcal{Y}) \triangleq \sup_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)| = \max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)|,$$

where $\mathcal{X}(A)$ denotes the probability of $\mathcal{X}$ to be in A , and similarly for $\mathcal{Y}(A)$. The second equation holds because we only consider Ω being a finite set in this problem.

(ii) Show that, if $\mathcal{X} \approx_{t,\varepsilon} \mathcal{Y}$ holds for all t , then $\Delta(\mathcal{X}, \mathcal{Y}) \leq \varepsilon$. In this case, we write $\mathcal{X} \approx_s \mathcal{Y}$.

(iii) Show that for two distribution $\mathcal{X}, \mathcal{Y}$ where $\Delta(\mathcal{X}, \mathcal{Y}) < \varepsilon$, $\mathcal{X} \approx_{t,\varepsilon} \mathcal{Y}$ holds for any t .

Answer (i).

Goal.

$$\max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)| = \frac{1}{2} \sum_{\omega \in \Omega} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)|$$

For every event $A \subseteq \Omega$, by triangle inequality we have

$$\begin{aligned} \frac{1}{2} \sum_{\omega \in \Omega} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)| &= \frac{1}{2} \left(\sum_{\omega \in A} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)| + \sum_{\omega \in \bar{A}} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)| \right) \\ &\stackrel{\text{🔴}}{\geq} \frac{1}{2} \left(\left| \sum_{\omega \in A} (\mathcal{X}(\omega) - \mathcal{Y}(\omega)) \right| + \left| \sum_{\omega \in \bar{A}} (\mathcal{X}(\omega) - \mathcal{Y}(\omega)) \right| \right) \quad (\text{by triangle inequality}) \\ &= \frac{1}{2} (|\mathcal{X}(A) - \mathcal{Y}(A)| + |\mathcal{X}(\bar{A}) - \mathcal{Y}(\bar{A})|) \\ &= \frac{1}{2} (|\mathcal{X}(A) - \mathcal{Y}(A)| + |(1 - \mathcal{X}(A)) - (1 - \mathcal{Y}(A))|) \\ &= \frac{1}{2} (|\mathcal{X}(A) - \mathcal{Y}(A)| + |\mathcal{X}(A) - \mathcal{Y}(A)|) \\ &= |\mathcal{X}(A) - \mathcal{Y}(A)|, \end{aligned}$$

so

$$\max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)| \leq \frac{1}{2} \sum_{\omega \in \Omega} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)|. \quad (1.1)$$

Notice that in the step (🔴) where we applied triangle inequality, the equality holds iff all $\mathcal{X}(\omega) - \mathcal{Y}(\omega)$ share the same sign (some possibly being 0) for $\omega \in A$ and all $\mathcal{X}(\omega) - \mathcal{Y}(\omega)$ share the same sign (some possibly being 0) for $\omega \in \bar{A}$. Therefore, the event

$$A^* \triangleq \{\omega \in \Omega \mid \mathcal{X}(\omega) \geq \mathcal{Y}(\omega)\} \subseteq \Omega \quad (1.2)$$

is a maximizer of $|\mathcal{X}(A) - \mathcal{Y}(A)|$. (One may check by definition that $\mathcal{X}(\omega) - \mathcal{Y}(\omega) \geq 0$ for $\omega \in A^*$ and $\mathcal{X}(\omega) - \mathcal{Y}(\omega) < 0$ for $\omega \in \overline{A^*}$.) To see this, observe that

$$\max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)| \geq |\mathcal{X}(A^*) - \mathcal{Y}(A^*)| \stackrel{\text{equality in (1.1) holds}}{=} \frac{1}{2} \sum_{\omega \in \Omega} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)| \stackrel{\text{by (1.1)}}{\geq} \max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)|,$$

which collapses into

$$\max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)| = |\mathcal{X}(A^*) - \mathcal{Y}(A^*)| = \frac{1}{2} \sum_{\omega \in \Omega} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)| \triangleq \Delta(\mathcal{X}, \mathcal{Y}), \quad (1.3)$$

as desired.

⊗

Answer (ii). We prove this by contradiction. Suppose to the contrary that $\Delta(\mathcal{X}, \mathcal{Y}) > \varepsilon$. We construct as follows a probabilistic distinguisher $\mathcal{D}$ with bounded running time that distinguishes $\mathcal{X}$ and $\mathcal{Y}$ with advantage greater than ε .

On input $\omega \in \Omega$, return 0 if $T(\omega) = 1$ and 1 otherwise, where $T : \Omega \rightarrow \{0, 1\}$ is an $|\Omega| \times 1$ bit table preconstructed by $T(\omega') := [\mathcal{X}(\omega') \geq \mathcal{Y}(\omega')]$ for every $\omega' \in \Omega$.

Since the preconstruction time for T is not counted in the running time of $\mathcal{D}$, $\mathcal{D}$ runs in only $O(|\Omega|)$ time, which is bounded. (Note that in the worst case $\mathcal{D}$ has to iterate the table T , which has size $|\Omega|$.) Moreover, the advantage of $\mathcal{D}$ is indeed greater than ε since

$$\begin{aligned} \left| \Pr_{\omega \leftarrow \mathcal{X}} [\mathcal{D}(\omega) = 1] - \Pr_{\omega \leftarrow \mathcal{Y}} [\mathcal{D}(\omega) = 1] \right| &= \left| \Pr_{\omega \leftarrow \mathcal{X}} [\mathcal{X}(\omega) < \mathcal{Y}(\omega)] - \Pr_{\omega \leftarrow \mathcal{Y}} [\mathcal{X}(\omega) < \mathcal{Y}(\omega)] \right| \\ &= \left| \Pr_{\omega \leftarrow \mathcal{X}} [\omega \in \overline{A^*}] - \Pr_{\omega \leftarrow \mathcal{Y}} [\omega \in \overline{A^*}] \right| && \text{(by definition of } A^*) \\ &= |\mathcal{X}(\overline{A^*}) - \mathcal{Y}(\overline{A^*})| \\ &= \frac{1}{2} \sum_{\omega \in \Omega} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)| && \text{(by (1.3))} \\ &\triangleq \Delta(\mathcal{X}, \mathcal{Y}) > \varepsilon, \end{aligned}$$

where A^* is the event defined in (1.2).

Hence, by definition, $\mathcal{X} \not\approx_{t, \varepsilon} \mathcal{Y}$ for some $t \in \mathbb{N}$, contradicting the assumption. ⊗

Answer (iii). We prove this by contradiction. Suppose to the contrary that $\mathcal{X} \not\approx_{t, \varepsilon} \mathcal{Y}$ for some $t \in \mathbb{N}$, which means that there exists a probabilistic distinguisher $\mathcal{D}$ running in time t such that

$$\left| \Pr_{\omega \leftarrow \mathcal{X}} [\mathcal{D}(\omega) = 1] - \Pr_{\omega \leftarrow \mathcal{Y}} [\mathcal{D}(\omega) = 1] \right| > \varepsilon.$$

Let $r \leftarrow \mathcal{Z}$ denote the internal randomness (or rather, random coin) of $\mathcal{D}$ for some distribution $\mathcal{Z}$ over $\{0, 1\}^\ell$, which allows us to write $\mathcal{D}$ as a deterministic function $\tilde{\mathcal{D}} : \Omega \times \{0, 1\}^\ell \rightarrow \{0, 1\}$:

$$\mathcal{D}(\omega) \equiv \tilde{\mathcal{D}}(\omega; r) \quad \text{where } r \stackrel{\$}{\leftarrow} \{0, 1\}^\ell.$$

To arrive at a contradiction, consider the events $A_r^\star \subseteq \Omega$ defined by

$$A_r^\star \triangleq \left\{ \omega \in \Omega \mid \tilde{\mathcal{D}}(\omega; r) = 1 \right\}$$

for every $r \in \{0, 1\}^\ell$. We then deduce that

$$\begin{aligned}
\varepsilon &< \left| \Pr_{\omega \leftarrow \mathcal{X}}[\mathcal{D}(\omega) = 1] - \Pr_{\omega \leftarrow \mathcal{Y}}[\mathcal{D}(\omega) = 1] \right| \\
&= \left| \Pr_{\omega \leftarrow \mathcal{X}, r \leftarrow \{0, 1\}^\ell}[\tilde{\mathcal{D}}(\omega; r) = 1] - \Pr_{\omega \leftarrow \mathcal{Y}, r \leftarrow \{0, 1\}^\ell}[\tilde{\mathcal{D}}(\omega; r) = 1] \right| \\
&= \left| \Pr_{\omega \leftarrow \mathcal{X}, r \leftarrow \{0, 1\}^\ell}[\omega \in A_r^\star] - \Pr_{\omega \leftarrow \mathcal{Y}, r \leftarrow \{0, 1\}^\ell}[\omega \in A_r^\star] \right| && \text{(by definition of } A_r^\star \text{)} \\
&= \left| \sum_{r \in \{0, 1\}^\ell} \Pr_{r' \leftarrow \mathcal{Z}}[r' = r] \Pr_{\omega \leftarrow \mathcal{X}}[\omega \in A_r^\star] - \sum_{r \in \{0, 1\}^\ell} \Pr_{r' \leftarrow \mathcal{Z}}[r' = r] \Pr_{\omega \leftarrow \mathcal{Y}}[\omega \in A_r^\star] \right| && \text{(by disjoint events and cond. prob.)} \\
&= \left| \sum_{r \in \{0, 1\}^\ell} \mathcal{Z}(r) \Pr_{\omega \leftarrow \mathcal{X}}[\omega \in A_r^\star] - \sum_{r \in \{0, 1\}^\ell} \mathcal{Z}(r) \Pr_{\omega \leftarrow \mathcal{Y}}[\omega \in A_r^\star] \right| \\
&= \left| \sum_{r \in \{0, 1\}^\ell} \mathcal{Z}(r) \left(\Pr_{\omega \leftarrow \mathcal{X}}[\omega \in A_r^\star] - \Pr_{\omega \leftarrow \mathcal{Y}}[\omega \in A_r^\star] \right) \right| \\
&\leq \sum_{r \in \{0, 1\}^\ell} \mathcal{Z}(r) \left| \Pr_{\omega \leftarrow \mathcal{X}}[\omega \in A_r^\star] - \Pr_{\omega \leftarrow \mathcal{Y}}[\omega \in A_r^\star] \right| && \text{(by triangle inequality)} \\
&= \sum_{r \in \{0, 1\}^\ell} \mathcal{Z}(r) |\mathcal{X}(A_r^\star) - \mathcal{Y}(A_r^\star)| \\
&\leq \sum_{r \in \{0, 1\}^\ell} \mathcal{Z}(r) \max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)| \\
&= \max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)| \cdot 1 \\
&= \Delta(\mathcal{X}, \mathcal{Y}), && \text{(by (1.3))}
\end{aligned}$$

which contradicts the assumption that $\Delta(\mathcal{X}, \mathcal{Y}) < \varepsilon$. Hence proved. $\otimes$

Problem 2 (Leftover Hash Lemma).

In this problem, you are asked to prove the leftover hash lemma you saw in the class. Recall the statement of the lemma as follows.

Theorem 2.1 (Leftover Hash Lemma). Let $n, m, q \in \mathbb{N}$ where q is a prime, and $\varepsilon \in (0, 1)$ be parameters satisfying $m \geq n \log q + 2 \log(\frac{1}{\varepsilon}) + 1$. Let $A \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ be a uniformly random matrix over $\mathbb{Z}_q^{m \times n}$ and $r \xleftarrow{\$} \{0, 1\}^m$. Then the distribution $(A, r^T A)$ is ε -close to uniform. Namely, let A', u denote the uniform distribution over $\mathbb{Z}_q^{m \times n}, \mathbb{Z}_q^n$, respectively, we have

$$\Delta((A, r^T A), (A', u)) \leq \varepsilon.$$

Before proving the lemma, we discuss the interpretation of the lemma. Intuitively, $r \xleftarrow{\$} \{0, 1\}^m$ has m bits of “entropy”, and $u \xleftarrow{\$} \mathbb{Z}_q^n$ has $n \log q$ bits of “entropy”. The lemma says that the uniformly random matrix $A \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ can be viewed as a *randomness extractor* that extracts the randomness of r and turn it into a close-to-uniform random variable $r^T A$, provided that r has sufficient entropy (i.e., m is sufficiently large). Now, you are asked to prove this lemma in the following four steps.

- (i) Show that for any $x, y \in \mathbb{Z}_q^m$ such that $x \neq y$,

$$\Pr_{A \xleftarrow{\$} \mathbb{Z}_q^{m \times n}} [x^T A = y^T A] \leq \frac{1}{q^n}.$$

- (ii) For a discrete random variable X , define the *collision probability* of X to be the probability that two independent samples of X taking the same value (i.e., they collide). Namely, define

$$\text{cp}(X) \triangleq \Pr[X = X'] = \sum_x \Pr[X = x]^2,$$

where X' denotes an independent copy of X . Use (i) to prove the following upper bound on the collision probability of $(A, r^T A)$:

$$\text{cp}(A, r^T A) \leq \frac{1}{q^{mn}} \left(\frac{1}{2^m} + \frac{1}{q^n} \right).$$

- (iii) Let X be a random variable with support size N . It is known that if X has collision probability $\text{cp}(X) \leq \frac{1+\varepsilon^2}{N}$, then X is ε -close to the uniform distribution over its support. Use this fact and (ii) to prove the lemma.

- (iv) Prove the above stated fact: if X has collision probability $\text{cp}(X) \leq \frac{1+\varepsilon^2}{N}$, then X is ε -close to uniform.

Answer (i). Let $x, y \in \mathbb{Z}_q^m$ be such that $x \neq y$. First, we simplify the goal:

$$\begin{aligned}
\Pr_{A \leftarrow \mathbb{Z}_q^{m \times n}} [x^T A = y^T A] &= \Pr_{A \leftarrow \mathbb{Z}_q^{m \times n}} [A^T(x - y) = \mathbf{0}] \\
&= \Pr_{A' \leftarrow \mathbb{Z}_q^{n \times m}} [A'(x - y) = \mathbf{0}] \quad (A' \triangleq A^T) \\
&= \Pr_{A' \leftarrow \mathbb{Z}_q^{n \times m}} \left[\sum_{j=1}^m A'_{1,j}(x - y)_j = 0 \wedge \cdots \wedge \sum_{j=1}^m A'_{n,j}(x - y)_j = 0 \right] \\
&\stackrel{(*)}{=} \prod_{i=1}^n \Pr_{A' \leftarrow \mathbb{Z}_q^{n \times m}} \left[\sum_{j=1}^m A'_{i,j}(x - y)_j = 0 \right] \quad (\text{by independence}) \\
&= \prod_{i=1}^n \Pr_{\alpha_1, \dots, \alpha_m \leftarrow \mathbb{Z}_q} \left[\sum_{j=1}^m \alpha_j(x - y)_j = 0 \right] \quad (\alpha_j \triangleq A'_{i,j} \ \forall j \in [m]) \\
&= \left(\Pr_{\alpha_1, \dots, \alpha_m \leftarrow \mathbb{Z}_q} \left[\sum_{j=1}^m \alpha_j(x - y)_j = 0 \right] \right)^n,
\end{aligned}$$

where $(*)$ holds because all entries of A' , i.e. $\{A'_{i,j}\}_{i \in [n], j \in [m]}$, are mutually independent, and so are $\{\sum_{j=1}^m A'_{i,j}(x - y)_j\}_{i \in [n], j \in [m]}$. If we can show that

$$\Pr_{\alpha_1, \dots, \alpha_m \leftarrow \mathbb{Z}_q} \left[\sum_{j=1}^m \alpha_j(x - y)_j = 0 \right] \leq \frac{1}{q}, \quad (2.1)$$

then we are done. Thus, it suffices to show (2.1) holds.

Goal. $\Pr_{\alpha_1, \dots, \alpha_m \leftarrow \mathbb{Z}_q} \left[\sum_{j=1}^m \alpha_j(x - y)_j = 0 \right] \leq \frac{1}{q}$

To show this, first notice that since $x \neq y$ and thus $x - y \neq \mathbf{0}$, there exists $t \in [m]$ such that $(x - y)_t \neq 0$.

Idea. α_t uniform and independent $\implies \alpha_t(x - y)_t$ uniform and indep. of other $\alpha_j(x - y)_j \implies \sum_{j=1}^m \alpha_j(x - y)_j$ uniform

For every $\beta \in \mathbb{Z}_q$, since $(x - y)_t \neq 0$ and thus $(x - y)_t^{-1}$ exists, we have

$$\Pr_{\alpha_t \leftarrow \mathbb{Z}_q} [\alpha_t(x - y)_t = \beta] = \Pr_{\alpha_t \leftarrow \mathbb{Z}_q} [\alpha_t = \beta(x - y)_t^{-1}] = \frac{1}{q}$$

as α_t is uniformly random. This means that the random variable $\alpha_t(x - y)_t \in \mathbb{Z}_q$ is also uniformly random.

Moreover, since $\alpha_1, \dots, \alpha_m$ are mutually independent and $(x - y)_1, \dots, (x - y)_m$ are just constants, we know $\alpha_t(x - y)_t \in \mathbb{Z}_q$ is independent of $\sum_{j \in [m] \setminus \{t\}} \alpha_j(x - y)_j \in \mathbb{Z}_q$ (the rest of sum). Therefore, as $\mathbb{Z}_q$ forms an additive group (of order q), by Problem 1-(ii) in HW1, their sum

$$\alpha_t(x - y)_t + \sum_{j \in [m] \setminus \{t\}} \alpha_j(x - y)_j = \sum_{j=1}^m \alpha_j(x - y)_j$$

is uniformly random over $\mathbb{Z}_q$. Hence, there is only $\frac{1}{q}$ probability that it is exactly 0, that is,

$$\Pr_{\alpha_1, \dots, \alpha_m \leftarrow \mathbb{Z}_q} \left[\sum_{j=1}^m \alpha_j(x - y)_j = 0 \right] = \frac{1}{q}, \text{ as claimed.} \quad \circledast$$

Answer (ii). Note that from (i), for all distinct $x, y \in \mathbb{Z}_q^m$ we have

$$\frac{1}{q^n} \geq \Pr_{A \leftarrow \mathbb{Z}_q^{m \times n}} [x^T A = y^T A] = \frac{|\{A \in \mathbb{Z}_q^{m \times n} \mid x^T A = y^T A\}|}{q^{mn}} = \sum_{A \in \mathbb{Z}_q^{m \times n}} \frac{1}{q^{mn}} [x^T A = y^T A]. \quad (2.2)$$

Hence,

$$\begin{aligned} \text{cp}(A, r^T A) &= \Pr_{\substack{A^{(1)}, A^{(2)} \leftarrow \mathbb{Z}_q^{m \times n} \\ r^{(1)}, r^{(2)} \leftarrow \{0,1\}^m}} [(A^{(1)}, r^{(1)T} A^{(1)}) = (A^{(2)}, r^{(2)T} A^{(2)})] \\ &= \Pr_{\substack{A^{(1)}, A^{(2)} \leftarrow \mathbb{Z}_q^{m \times n} \\ r^{(1)}, r^{(2)} \leftarrow \{0,1\}^m}} [A^{(1)} = A^{(2)} \wedge r^{(1)T} A^{(1)} = r^{(2)T} A^{(2)}] \\ &= \sum_{\substack{A' \in \mathbb{Z}_q^{m \times n} \\ r' \in \{0,1\}^m}} \Pr_{A^{(2)} \leftarrow \mathbb{Z}_q^{m \times n}} [A^{(2)} = A'] \Pr_{r^{(2)} \leftarrow \{0,1\}^m} [r^{(2)} = r'] \Pr_{\substack{A \leftarrow \mathbb{Z}_q^{m \times n} \\ r \leftarrow \{0,1\}^m}} [A^{(1)} = A' \wedge r^{(1)T} A^{(1)} = r'^T A'] \\ &= \sum_{\substack{A' \in \mathbb{Z}_q^{m \times n} \\ r' \in \{0,1\}^m}} \frac{1}{q^{mn}} \frac{1}{2^m} \Pr_{A \leftarrow \mathbb{Z}_q^{m \times n}, r \leftarrow \{0,1\}^m} [A = A' \wedge r^T A = r'^T A'] \\ &= \frac{1}{q^{mn}} \frac{1}{2^m} \sum_{A' \in \mathbb{Z}_q^{m \times n}} \sum_{r' \in \{0,1\}^m} \Pr_{A \leftarrow \mathbb{Z}_q^{m \times n}, r \leftarrow \{0,1\}^m} [A = A' \wedge r^T A = r'^T A'] \\ &= \frac{1}{q^{mn}} \frac{1}{2^m} \sum_{A' \in \mathbb{Z}_q^{m \times n}} \sum_{r' \in \{0,1\}^m} \Pr_{A \leftarrow \mathbb{Z}_q^{m \times n}} [A = A'] \Pr_{r \leftarrow \{0,1\}^m} [r^T A' = r'^T A'] \quad (\because r^T A' \perp\!\!\!\perp A) \\ &= \frac{1}{q^{2mn}} \frac{1}{2^m} \sum_{A' \in \mathbb{Z}_q^{m \times n}} \sum_{r' \in \{0,1\}^m} \underbrace{\Pr_{r \leftarrow \{0,1\}^m} [r^T A' = r'^T A']}_{= \frac{|\{r \in \{0,1\}^m \mid r^T A' = r'^T A'\}|}{2^m}} \\ &= \frac{1}{q^{2mn}} \frac{1}{2^m} \sum_{A' \in \mathbb{Z}_q^{m \times n}} \sum_{r' \in \{0,1\}^m} \overbrace{\sum_{r \in \{0,1\}^m} \frac{1}{2^m} [r^T A' = r'^T A']} \\ &= \frac{1}{q^{2mn}} \frac{1}{2^{2m}} \sum_{A' \in \mathbb{Z}_q^{m \times n}} \sum_{r' \in \{0,1\}^m} \sum_{r \in \{0,1\}^m} [r^T A' = r'^T A'] \\ &= \frac{1}{q^{2mn}} \frac{1}{2^{2m}} \sum_{A' \in \mathbb{Z}_q^{m \times n}} \sum_{r' \in \{0,1\}^m} \left(1 + \sum_{r \in \{0,1\}^m, r \neq r'} [r^T A' = r'^T A'] \right) \quad (\because r^T A' = r'^T A' \text{ when } r = r') \\ &= \frac{1}{q^{mn}} \frac{1}{2^m} + \frac{1}{q^{2mn}} \frac{1}{2^{2m}} \sum_{r' \in \{0,1\}^m} \sum_{r \in \{0,1\}^m, r \neq r'} \sum_{A' \in \mathbb{Z}_q^{m \times n}} [r^T A' = r'^T A'] \\ &= \frac{1}{q^{mn}} \frac{1}{2^m} + \frac{1}{q^{mn}} \frac{1}{2^{2m}} \sum_{r' \in \{0,1\}^m} \sum_{r \in \{0,1\}^m, r \neq r'} \underbrace{\sum_{A' \in \mathbb{Z}_q^{m \times n}} \frac{1}{q^{mn}} [r^T A' = r'^T A']}_{= \Pr_{A \leftarrow \mathbb{Z}_q^{m \times n}} [r^T A = r'^T A] \leq \frac{1}{q^n}} \\ &\quad \text{(by (2.2))} \\ &\leq \frac{1}{q^{mn}} \left(\frac{1}{2^m} + \frac{2^m(2^m - 1)}{2^{2m}} \cdot \frac{1}{q^n} \right) \leq \frac{1}{q^{mn}} \left(\frac{1}{2^m} + 1 \cdot \frac{1}{q^n} \right) = \frac{1}{q^{mn}} \left(\frac{1}{2^m} + \frac{1}{q^n} \right), \end{aligned}$$

as desired.

Answer (iii). Since we have the fact

Lemma 2.1 (Fact). Let X be a random variable. Then

$$\text{cp}(X) \leq \frac{1 + \varepsilon^2}{|\text{Supp}(X)|} \implies \Delta(X, U_{\text{Supp}(X)}) \leq \varepsilon.$$

to prove

$$\Delta((A, r^T A), (A, U_{\mathbb{Z}_q^n})) \leq \varepsilon,$$

it suffices to show that

$$\text{cp}(A, r^T A) \leq \frac{1 + \varepsilon^2}{|\mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^n|} = \frac{1 + \varepsilon^2}{q^{mn+n}}.$$

To show this, we simply use the result of (ii) and the assumption that $m \geq n \log q + 2 \log(\frac{1}{\varepsilon}) + 1$:

$$\text{cp}(A, r^T A) \stackrel{\text{by (ii)}}{\leq} \frac{1}{q^{mn}} \left(\frac{1}{2^m} + \frac{1}{q^n} \right) \leq \frac{1}{q^{mn}} \left(\frac{1}{q^n \cdot \frac{1}{\varepsilon^2} \cdot 2} + \frac{1}{q^n} \right) < \frac{1}{q^{mn}} \left(\frac{\varepsilon^2}{q^n} + \frac{1}{q^n} \right) = \frac{1 + \varepsilon^2}{q^{mn+n}}.$$

Answer (iv). To prove **Lemma 2.1**, let X be a random variable over Ω , where $|\Omega| = N$, and assume that $\text{cp}(X) \leq \frac{1 + \varepsilon^2}{N}$. Our goal is to show that $\Delta(X, U_\Omega) \stackrel{\text{def}}{=} \frac{1}{2} \sum_{\omega \in \Omega} |\Pr[X = \omega] - \frac{1}{N}| \leq \varepsilon$, where U_Ω denotes the uniform distribution over Ω . (Recall from Problem 1 the definition of $\Delta(\cdot, \cdot)$.)

Idea. As $\Delta(X, U_\Omega) \triangleq \frac{1}{2} \sum_{\omega \in \Omega} |\Pr[X = \omega] - \frac{1}{N}|$ is basically the sum of $|\Pr[X = \omega] - \frac{1}{N}|$, we try to express $\text{cp}(X) = \sum_{\omega \in \Omega} \Pr[X = \omega]^2$ in terms of $|\Pr[X = \omega] - \frac{1}{N}|$.

Let $A \subseteq \Omega$ denote the event $\{\omega \in \Omega \mid \Pr[X = \omega] \geq \frac{1}{N}\}$. Using A , we rewrite $\text{cp}(X)$ as follows:

$$\begin{aligned} \frac{1 + \varepsilon^2}{N} &\geq \text{cp}(X) \triangleq \Pr[X = X'] = \sum_{\omega \in \Omega} \Pr[X = X' = \omega] \\ &= \sum_{\omega \in \Omega} \Pr[X = \omega] \Pr[X' = \omega] && (\because X \perp\!\!\!\perp X') \\ &= \sum_{\omega \in \Omega} \Pr[X = \omega]^2 \\ &= \sum_{\omega \in A} \left(\left| \Pr[X = \omega] - \frac{1}{N} \right| + \frac{1}{N} \right)^2 + \sum_{\omega \in \bar{A}} \left(\left| \Pr[X = \omega] - \frac{1}{N} \right| - \frac{1}{N} \right)^2 \\ &= \sum_{\omega \in \Omega} \left| \Pr[X = \omega] - \frac{1}{N} \right|^2 + \underbrace{\frac{2}{N} \sum_{\omega \in A} \left| \Pr[X = \omega] - \frac{1}{N} \right|}_{= \Pr[X \in A] - \frac{|A|}{N}} - \underbrace{\frac{2}{N} \sum_{\omega \in \bar{A}} \left| \Pr[X = \omega] - \frac{1}{N} \right|}_{= \frac{|\bar{A}|}{N} - \Pr[X \in \bar{A}]} + \frac{1}{N} \\ &= \sum_{\omega \in \Omega} \left| \Pr[X = \omega] - \frac{1}{N} \right|^2 + \frac{1}{N}, \end{aligned}$$

which implies

$$\sum_{\omega \in \Omega} \left| \Pr[X = \omega] - \frac{1}{N} \right|^2 \leq \frac{\varepsilon^2}{N}. \quad (2.3)$$

Finally, by Cauchy–Schwarz inequality, we have

$$\left(\sum_{\omega \in \Omega} \left| \Pr[X = \omega] - \frac{1}{N} \right| \right)^2 = \left(\sum_{\omega \in \Omega} \left| \Pr[X = \omega] - \frac{1}{N} \right| \cdot 1 \right)^2 \leq \left(\sum_{\omega \in \Omega} \left| \Pr[X = \omega] - \frac{1}{N} \right|^2 \right) \left(\sum_{\omega \in \Omega} 1^2 \right) \stackrel{(2.3)}{\leq} \left(\frac{\varepsilon^2}{N} \right) \cdot N = \varepsilon^2,$$

and therefore

$$\Delta(X, U_\Omega) \stackrel{\text{def}}{=} \frac{1}{2} \sum_{\omega \in \Omega} \left| \Pr[X = \omega] - \frac{1}{N} \right| \leq \frac{1}{2} \varepsilon < \varepsilon.$$

(Note that $\sum_{\omega \in \Omega} \left| \Pr[X = \omega] - \frac{1}{N} \right|$ is nonnegative.)

⊛

Problem 3 (Fun with the Equivalent CPA definitions).

Show that the following security definitions for PKE (defined by the following security games) are equivalent to CPA-security or not. Prove your answer or construct a counter-example.

- (i) $\text{PKE}^{\text{ZeroEnc}}$
- (ii) $\text{PKE}^{\text{VectorEnc}}$
- (iii) $\text{PKE}^{\text{Same-or-not}}$
- (iv) $\text{PKE}^{\text{Rand-Same-or-not}}$

Answer (i). We show that ZeroEnc security is equivalent to IND-CPA security for all PKE schemes.

Note. ZeroEnc security looks harder to break than IND-CPA security. However, it turns out that they are equivalently hard to break.

IND-CPA security $\implies$ ZeroEnc security.

We prove this by contraposition. Let $(\text{KeyGen}, \text{Enc}, \text{Dec})$ be a non-ZeroEnc-secure PKE scheme. Then by definition, there exist a PPT adversary Adv and a nonnegligible function ε such that for every $n \in \mathbb{N}$, in the experiment $\text{PKE}^{\text{ZeroEnc}}$ we have

$$\Pr[\text{Adv wins}] = \frac{1}{2} + \varepsilon(n). \quad (3.1)$$

We then construct a PPT reduction R that breaks the IND-CPA security of $(\text{KeyGen}, \text{Enc}, \text{Dec})$ with nonnegligible advantage, as illustrated in the following experiment.

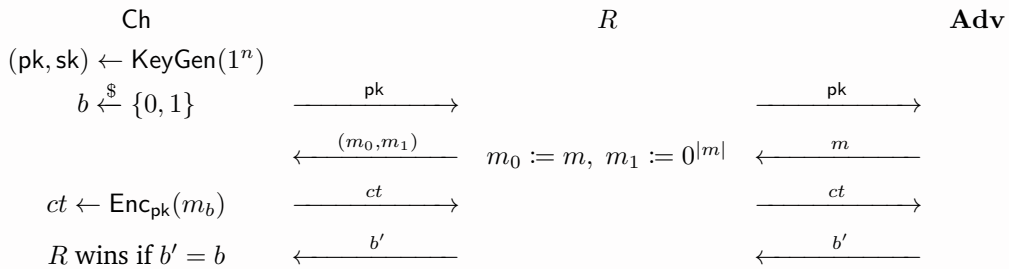


Figure 1: Experiment $\text{PKE}^{\text{IND-CPA}}$

Clearly R is a PPT algorithm since Adv is a PPT algorithm. Also, it's straightforward that in this experiment, the ciphertext ct given by R to Adv is distributed exactly as in the experiment $\text{PKE}^{\text{ZeroEnc}}$. Hence R wins here iff Adv wins in $\text{PKE}^{\text{ZeroEnc}}$. This implies that for all $n \in \mathbb{N}$

$$\Pr[R \text{ wins in } \text{PKE}^{\text{IND-CPA}}] = \Pr[\text{Adv wins in } \text{PKE}^{\text{ZeroEnc}}] \stackrel{(3.1)}{=} \frac{1}{2} + \varepsilon(n),$$

meaning that R wins in $\text{PKE}^{\text{IND-CPA}}$ with nonnegligible advantage. Therefore, by definition, $(\text{KeyGen}, \text{Enc}, \text{Dec})$ not IND-CPA-secure.

IND-CPA security $\Leftarrow$ ZeroEnc security.

Likewise, we prove this by contraposition. Let $(\text{KeyGen}, \text{Enc}, \text{Dec})$ be a non-IND-CPA-secure PKE scheme. Then by definition, there exist a PPT adversary Adv and a nonnegligible function ε such

that for every $n \in \mathbb{N}$, in the experiment $\text{PKE}^{IND-CPA}$ we have

$$\Pr[\mathbf{Adv} \text{ wins}] = \frac{1}{2} + \varepsilon(n). \quad (3.2)$$

We then construct a PPT reduction R that breaks the ZeroEnc security of $(\text{KeyGen}, \text{Enc}, \text{Dec})$ with nonnegligible advantage, as illustrated in the following experiment.

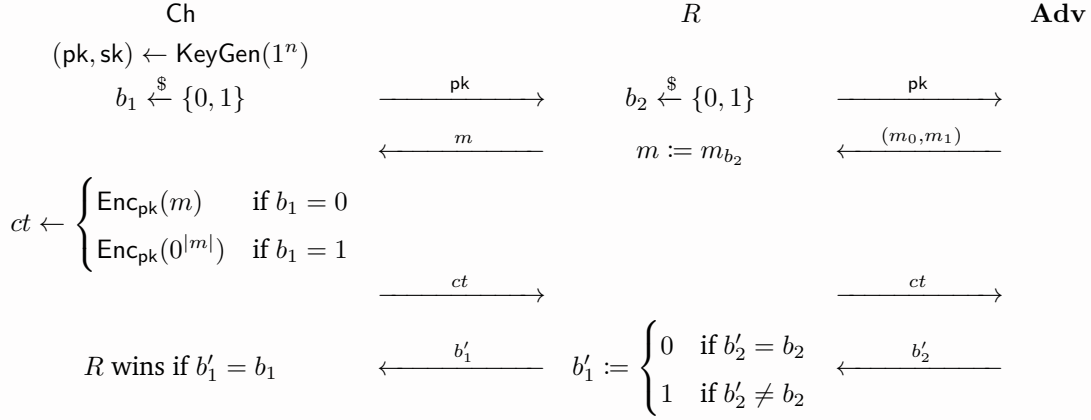


Figure 2: Experiment $\text{PKE}^{ZeroEnc}$

Clearly R is a PPT algorithm since $\mathbf{Adv}$ is a PPT algorithm. To analyze the win rate of R , observe that for all $n \in \mathbb{N}$:

- If $b_1 = 0$, then $ct \leftarrow \text{Enc}_{pk}(m_{b_2})$, meaning that the ciphertext ct given by R to $\mathbf{Adv}$ is distributed identically to ct in the experiment $\text{PKE}^{IND-CPA}$. Consequently, R wins here iff $\mathbf{Adv}$ correctly guesses b_2 , whose probability is equal to the probability that $\mathbf{Adv}$ wins in $\text{PKE}^{IND-CPA}$, that is,

$$\Pr_{b_1 \xleftarrow{\$} \{0,1\}} [R \text{ wins} \mid b_1 = 0] = \Pr_{b_1, b_2 \xleftarrow{\$} \{0,1\}} [b'_2 = b_2 \mid b_1 = 0] = \Pr_{b_1, b_2 \xleftarrow{\$} \{0,1\}} [\mathbf{Adv} \text{ wins in } \text{PKE}^{IND-CPA}] \stackrel{(3.2)}{=} \frac{1}{2} + \varepsilon(n).$$

- If $d = 1$, then $ct \leftarrow \text{Enc}_{pk}(0^{|m|})$, then we see that the ciphertext ct given by R to $\mathbf{Adv}$ is independent of b_2 , and so does the output b'_2 of $\mathbf{Adv}$. Since $b_2 \xleftarrow{\$} \{0, 1\}$ is uniformly random, this implies that $\Pr[b'_2 = \overline{b_2} \mid b_1 = 1] = \frac{1}{2}$. Intuitively speaking, ct leaks to $\mathbf{Adv}$ zero information of the bit b_2 , so $\mathbf{Adv}$ correctly guesses the value of b_2 with probability only $\frac{1}{2}$. In other words,

$$\Pr_{b_1 \xleftarrow{\$} \{0,1\}} [R \text{ wins} \mid b_1 = 1] = \Pr_{b_1, b_2 \xleftarrow{\$} \{0,1\}} [b'_2 = \overline{b_2} \mid b_1 = 1] = \Pr_{\substack{b_2 \xleftarrow{\$} \{0,1\} \\ (pk, sk) \leftarrow \text{KeyGen}(1^n)}} [\mathbf{Adv}(pk, \text{Enc}_{pk}(0^{|m|})) = \overline{b_2}] = \frac{1}{2}$$

since $b_2 \in \{0, 1\}$ is independent of LHS and is uniformly random.

Putting both together, we see that for all $n \in \mathbb{N}$

$$\Pr_{b_1 \xleftarrow{\$} \{0,1\}} [R \text{ wins}] = \frac{1}{2} \Pr_{b_1 \xleftarrow{\$} \{0,1\}} [R \text{ wins} \mid b_1 = 0] + \frac{1}{2} \Pr_{b_1 \xleftarrow{\$} \{0,1\}} [R \text{ wins} \mid b_1 = 1] = \frac{1}{2} \left(\frac{1}{2} + \varepsilon(n) \right) + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2} + \frac{\varepsilon(n)}{2},$$

where the advantage $\frac{\varepsilon(n)}{2}$ of R in $\text{PKE}^{ZeroEnc}$ is nonnegligible. Therefore, by definition, $(\text{KeyGen}, \text{Enc}, \text{Dec})$ not ZeroEnc-secure. ⊛

Answer (ii). We show that VectorEnc security is equivalent to IND-CPA security for all PKE schemes.

Note. VectorEnc security looks easier to break than IND-CPA security. However, it turns out that they are equivalently hard to break.

VectorEnc security $\implies$ IND-CPA security.

We prove this by contraposition. Let $(\text{KeyGen}, \text{Enc}, \text{Dec})$ be a non-IND-CPA-secure PKE scheme. Then by definition, there exist a PPT adversary Adv and a nonnegligible function ε such that for every $n \in \mathbb{N}$, in the experiment $\text{PKE}^{\text{IND-CPA}}$ we have

$$\Pr[\text{Adv wins}] = \frac{1}{2} + \varepsilon(n). \quad (3.3)$$

We then construct a PPT reduction R that breaks the VectorEnc security of $(\text{KeyGen}, \text{Enc}, \text{Dec})$ with nonnegligible advantage, as illustrated in the following experiment, where we suppose $0^n \in \mathcal{M}_n$.

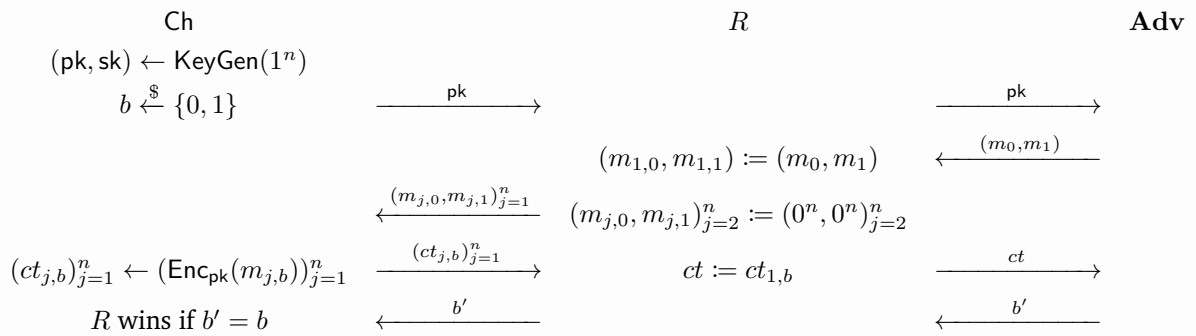


Figure 3: Experiment $\text{PKE}^{\text{VectorEnc}}$

Clearly R is a PPT algorithm since Adv is a PPT algorithm. Also, it's straightforward that in this experiment, the ciphertext $ct = ct_{1,b} \leftarrow \text{Enc}_{\text{pk}}(m_{1,b})$ given by R to Adv is distributed identically to ct in the experiment $\text{PKE}^{\text{IND-CPA}}$. Hence R wins here iff Adv wins in $\text{PKE}^{\text{IND-CPA}}$. This implies that for all $n \in \mathbb{N}$

$$\Pr[R \text{ wins in } \text{PKE}^{\text{VectorEnc}}] = \Pr[\text{Adv wins in } \text{PKE}^{\text{IND-CPA}}] \stackrel{(3.3)}{=} \frac{1}{2} + \varepsilon(n),$$

meaning that R wins in $\text{PKE}^{\text{VectorEnc}}$ with nonnegligible advantage. Therefore, by definition, $(\text{KeyGen}, \text{Enc}, \text{Dec})$ is not VectorEnc-secure.

IND-CPA security $\implies$ VectorEnc security.

We prove this by contraposition, with the use of hybrid argument. Let $(\text{KeyGen}, \text{Enc}, \text{Dec})$ be a non-VectorEnc-secure PKE scheme. Then by definition, there exist a PPT adversary Adv and a nonnegligible function ε such that for every $n \in \mathbb{N}$, in the experiment $\text{PKE}^{\text{VectorEnc}}$ we have

$$\Pr[\text{Adv wins}] = \frac{1}{2} + \varepsilon(n). \quad (3.4)$$

Expanding $\Pr[\mathbf{Adv} \text{ wins}]$, we see that

$$\begin{aligned}
\frac{1}{2} + \varepsilon(n) &= \Pr[\mathbf{Adv} \text{ wins}] \\
&= \frac{1}{2} \Pr \left[\begin{array}{l} (\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk}) \end{array} : \mathbf{Adv}(\mathbf{pk}, (\text{Enc}_{\mathbf{pk}}(m_{j,0}))_{j=1}^n) = 0 \right] \\
&+ \frac{1}{2} \Pr \left[\begin{array}{l} (\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk}) \end{array} : \mathbf{Adv}(\mathbf{pk}, (\text{Enc}_{\mathbf{pk}}(m_{j,1}))_{j=1}^n) = 1 \right] \\
&= \frac{1}{2} \left(1 - \Pr \left[\begin{array}{l} (\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk}) \end{array} : \mathbf{Adv}(\mathbf{pk}, (\text{Enc}_{\mathbf{pk}}(m_{j,0}))_{j=1}^n) = 1 \right] \right) \\
&+ \frac{1}{2} \Pr \left[\begin{array}{l} (\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk}) \end{array} : \mathbf{Adv}(\mathbf{pk}, (\text{Enc}_{\mathbf{pk}}(m_{j,1}))_{j=1}^n) = 1 \right] \\
&= \frac{1}{2} + \frac{1}{2} \left(\Pr \left[\begin{array}{l} (\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk}) \end{array} : \mathbf{Adv}(\mathbf{pk}, (\text{Enc}_{\mathbf{pk}}(m_{j,1}))_{j=1}^n) = 1 \right] \right. \\
&\quad \left. - \Pr \left[\begin{array}{l} (\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk}) \end{array} : \mathbf{Adv}(\mathbf{pk}, (\text{Enc}_{\mathbf{pk}}(m_{j,0}))_{j=1}^n) = 1 \right] \right),
\end{aligned}$$

so we have

$$\begin{aligned}
&\Pr \left[\begin{array}{l} (\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk}) \end{array} : \mathbf{Adv}(\mathbf{pk}, (\text{Enc}_{\mathbf{pk}}(m_{j,1}))_{j=1}^n) = 1 \right] \\
&- \Pr \left[\begin{array}{l} (\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk}) \end{array} : \mathbf{Adv}(\mathbf{pk}, (\text{Enc}_{\mathbf{pk}}(m_{j,0}))_{j=1}^n) = 1 \right] = 2\varepsilon(n). \quad (3.5)
\end{aligned}$$

We then construct a PPT reduction R that breaks the IND-CPA security of $(\text{KeyGen}, \text{Enc}, \text{Dec})$ with nonnegligible advantage, as illustrated in the following experiment.

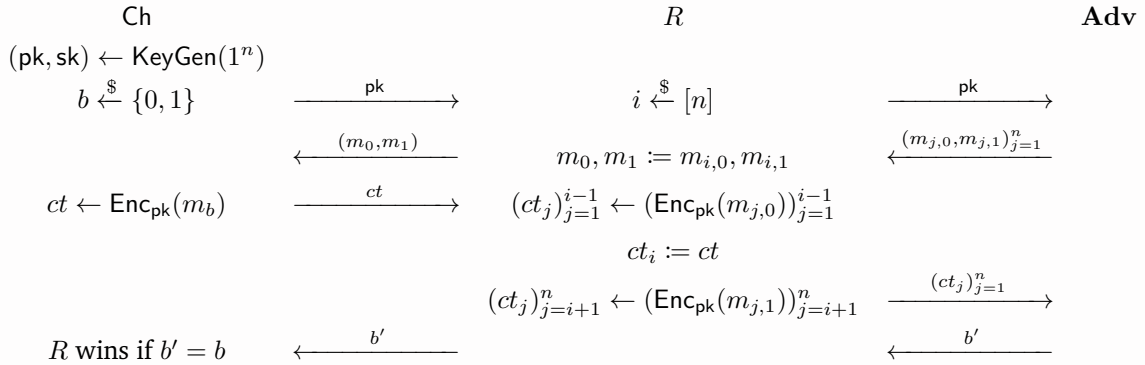


Figure 4: Experiment $\text{PKE}^{\text{IND-CPA}}$

Clearly R is a PPT algorithm since $\mathbf{Adv}$ is a PPT algorithm. To analyze the advantage of R in this experiment, for every $n \in \mathbb{N}$ and $i \in \{0, 1, \dots, n\}$, for random variables $(\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^n)$ and $(m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathbf{pk})$, we introduce the following hybrid distributions over $\mathcal{C}_n^n$:

$$\text{Hyb}_i \triangleq \left(\underbrace{\text{Enc}_{\mathbf{pk}}(m_{0,0}), \dots, \text{Enc}_{\mathbf{pk}}(m_{i,0})}_{\text{ciphertext of } (m_{j,0})_{j=1}^i}, \underbrace{\text{Enc}_{\mathbf{pk}}(m_{i+1,1}), \dots, \text{Enc}_{\mathbf{pk}}(m_{n,1})}_{\text{ciphertext of } (m_{j,1})_{j=i+1}^n} \right).$$

With the notion of hybrid distributions, we rewrite (3.5) as

$$\Pr_{(pk,sk) \leftarrow \text{KeyGen}(1^n)} [\text{Adv}(pk, \text{Hyb}_0)] - \Pr_{(pk,sk) \leftarrow \text{KeyGen}(1^n)} [\text{Adv}(pk, \text{Hyb}_n)] = 2\varepsilon(n). \quad (3.6)$$

Back to R participating in $\text{PKE}^{\text{IND-CPA}}$. It's now easy to see that for every $n \in \mathbb{N}$, the win rate of R is actually

$$\begin{aligned} \Pr[R \text{ wins}] &= \Pr \left[\begin{array}{c} i \xleftarrow{\$} [n] \\ (pk, sk) \leftarrow \text{KeyGen}(1^n) \\ b \xleftarrow{\$} \{0, 1\} \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \text{Adv}(pk) \end{array} : \text{Adv} \left(pk, \left(\begin{array}{c} \text{Enc}_{pk}(m_{1,0}), \dots, \text{Enc}_{pk}(m_{i-1,0}), \text{Enc}_{pk}(m_{i,b}), \\ \text{Enc}_{pk}(m_{i+1,1}), \dots, \text{Enc}_{pk}(m_{n,1}) \end{array} \right) \right) = 1 \right] \\ &= \frac{1}{2} \Pr \left[\begin{array}{c} i \xleftarrow{\$} [n] \\ (pk, sk) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \text{Adv}(pk) \end{array} : \text{Adv} \left(pk, \left(\begin{array}{c} \text{Enc}_{pk}(m_{1,0}), \dots, \text{Enc}_{pk}(m_{i,0}), \\ \text{Enc}_{pk}(m_{i+1,1}), \dots, \text{Enc}_{pk}(m_{n,1}) \end{array} \right) \right) = 0 \right] \\ &\quad + \frac{1}{2} \Pr \left[\begin{array}{c} i \xleftarrow{\$} [n] \\ (pk, sk) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \text{Adv}(pk) \end{array} : \text{Adv} \left(pk, \left(\begin{array}{c} \text{Enc}_{pk}(m_{1,0}), \dots, \text{Enc}_{pk}(m_{i-1,0}), \\ \text{Enc}_{pk}(m_{i,1}), \dots, \text{Enc}_{pk}(m_{n,1}) \end{array} \right) \right) = 1 \right] \\ &= \frac{1}{2} + \frac{1}{2} \left(\Pr \left[\begin{array}{c} i \xleftarrow{\$} [n] \\ (pk, sk) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \text{Adv}(pk) \end{array} : \text{Adv} \left(pk, \left(\begin{array}{c} \text{Enc}_{pk}(m_{1,0}), \dots, \text{Enc}_{pk}(m_{i-1,0}), \\ \text{Enc}_{pk}(m_{i,1}), \dots, \text{Enc}_{pk}(m_{n,1}) \end{array} \right) \right) = 1 \right] \right. \\ &\quad \left. - \Pr \left[\begin{array}{c} i \xleftarrow{\$} [n] \\ (pk, sk) \leftarrow \text{KeyGen}(1^n) \\ (m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \text{Adv}(pk) \end{array} : \text{Adv} \left(pk, \left(\begin{array}{c} \text{Enc}_{pk}(m_{1,0}), \dots, \text{Enc}_{pk}(m_{i,0}), \\ \text{Enc}_{pk}(m_{i+1,1}), \dots, \text{Enc}_{pk}(m_{n,1}) \end{array} \right) \right) = 0 \right] \right) \\ &= \frac{1}{2} + \frac{1}{2} \left(\Pr_{\substack{i \xleftarrow{\$} [n] \\ (pk,sk) \leftarrow \text{KeyGen}(1^n)}} [\text{Adv}(pk, \text{Hyb}_{i-1}) = 1] - \Pr_{\substack{i \xleftarrow{\$} [n] \\ (pk,sk) \leftarrow \text{KeyGen}(1^n)}} [\text{Adv}(pk, \text{Hyb}_i) = 1] \right) \\ &= \frac{1}{2} + \frac{1}{2} \left(\sum_{i \in [n]} \frac{1}{n} \Pr_{(pk,sk) \leftarrow \text{KeyGen}(1^n)} [\text{Adv}(pk, \text{Hyb}_{i-1}) = 1] - \sum_{i \in [n]} \frac{1}{n} \Pr_{(pk,sk) \leftarrow \text{KeyGen}(1^n)} [\text{Adv}(pk, \text{Hyb}_i) = 1] \right) \\ &= \frac{1}{2} + \frac{1}{2n} \left(\Pr_{(pk,sk) \leftarrow \text{KeyGen}(1^n)} [\text{Adv}(pk, \text{Hyb}_0) = 1] - \Pr_{(pk,sk) \leftarrow \text{KeyGen}(1^n)} [\text{Adv}(pk, \text{Hyb}_n) = 1] \right) \\ &= \frac{1}{2} + \frac{1}{2n} (2\varepsilon(n)) \quad (\text{by (3.6)}) \\ &= \frac{1}{2} + \frac{\varepsilon(n)}{n}. \end{aligned}$$

Therefore, for all $n \in \mathbb{N}$, the advantage $\frac{\varepsilon(n)}{n}$ of R is nonnegligible since ε is nonnegligible.

By definition, $(\text{KeyGen}, \text{Enc}, \text{Dec})$ is hence not IND-CPA secure. ⊗

Answer (iii). idk ⊗

Answer (iv). Assuming the existence of Rand-Same-or-not-secure PKE schemes (otherwise the implication is trivial), we show that Rand-Same-or-not security does not imply IND-CPA security by providing a counterexample.

Note. Rand-Same-or-not security is **strictly** harder to break than IND-CPA security. That is,

IND-CPA-secure $\implies$ Rand-Same-or-not-secure but IND-CPA-secure $\not\Leftarrow$ Rand-Same-or-not-secure.

Let $(\text{KeyGen}, \text{Enc}, \text{Dec})$ be a Rand-Same-or-not-secure PKE scheme with message space $\mathcal{M}_n$ and ciphertext space $\mathcal{C}_n$, where $|\mathcal{M}_n| = 2^n$. For every $n \in \mathbb{N}$, fix a message $m_n^* \in \mathcal{M}_n$ and a special ciphertext $ct_n^* \notin \mathcal{C}_n$ that is not in the ciphertext space. Define the new ciphertext space $\mathcal{C}'_n \triangleq \mathcal{C}_n \cup \{ct_n^*\}$, along with the new PPT encryption and decryption algorithms Enc' , Dec' by

$$\begin{aligned} \text{Enc}'_{\text{pk}}(m; r) &\triangleq \begin{cases} ct_n^* & \text{if } m = m_n^* \\ \text{Enc}_{\text{pk}}(m; r) & \text{otherwise} \end{cases} \quad \text{for } m \in \mathcal{M}_n \\ \text{Dec}'_{\text{sk}}(ct) &\triangleq \begin{cases} m_n^* & \text{if } ct = ct_n^* \\ \text{Dec}_{\text{sk}}(ct) & \text{otherwise} \end{cases} \quad \text{for } ct \in \mathcal{C}'_n = \mathcal{C}_n \cup \{ct_n^*\}, \end{aligned}$$

where pk, sk denote any public and private key generated by KeyGen and r denotes the internal randomness of Enc . (Clearly Enc' and Dec' run in PPT by constructions.) We claim that $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ is a PKE scheme with message space $\mathcal{M}_n$ and ciphertext space $\mathcal{C}'_n$ that is Rand-Same-or-not-secure but not IND-CPA-secure.

Claim 3.1. $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ is a PKE scheme with message space $\mathcal{M}_n$ and ciphertext space $\mathcal{C}'_n$.

Proof. Clearly Enc' and Dec' are PPT algorithms by construction, so $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ are all PPT algorithms. To show correctness, notice that by the correctness of $(\text{KeyGen}, \text{Enc}, \text{Dec})$, there exists $\mu(n) = \text{negl}(n)$ such that for every $n \in \mathbb{N}$ and $m \in \mathcal{M}_n \setminus \{m_n^*\}$,

$$\Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [\text{Dec}'_{\text{sk}}(\text{Enc}'_{\text{pk}}(m)) = m] = \Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [\text{Dec}_{\text{sk}}(\text{Enc}_{\text{pk}}(m)) = m] \geq 1 - \mu(n).$$

As for the correctness on $m = m_n^*$ for each $n \in \mathbb{N}$, this is given directly by the constructions of Enc' and Dec' :

$$\Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [\text{Dec}'_{\text{sk}}(\text{Enc}'_{\text{pk}}(m_n^*)) = m_n^*] = \Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [m_n^* = m_n^*] = 1 \geq 1 - \mu(n).$$

Hence, $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ is indeed a correct PKE scheme. $\blacksquare$

It remains to show that $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ is Rand-Same-or-not-secure but not IND-CPA-secure.

Claim 3.2. $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ is Rand-Same-or-not-secure.

Proof. Suppose to the contrary that $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ is not Rand-Same-or-not-secure. Then by definition, there exist a PPT adversary Adv and a nonnegligible function ε such that for every $n \in \mathbb{N}$, in the experiment $\text{PKE}^{\text{Rand-Same-or-not}}$ for $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ we have

$$\Pr_{\substack{m, m' \xleftarrow{\$} \mathcal{M}_n \\ b \xleftarrow{\$} \{0, 1\} \\ (\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)}} [\text{Adv} \left(\text{pk}, \begin{cases} \{\text{Enc}_{\text{pk}}(m), \text{Enc}_{\text{pk}}(m)\} & \text{if } b = 0 \\ \{\text{Enc}_{\text{pk}}(m), \text{Enc}_{\text{pk}}(m')\} & \text{if } b = 1 \end{cases} \right) = b] = \frac{1}{2} + \varepsilon(n). \quad (3.7)$$

$\blacksquare$

cont'd. We show that this **Adv** also breaks the Rand-Same-or-not security of (KeyGen, Enc, Dec) with nonnegligible advantage, as illustrated in the following experiment. To avoid ambiguity, we denote by R the **Adv** participating in the experiment for (KeyGen, Enc, Dec).

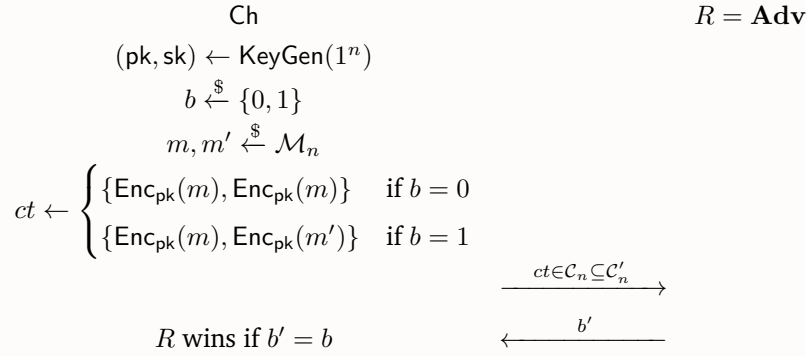


Figure 5: Experiment $\text{PKE}^{\text{Rand-Same-or-not}}$

Note that $R \triangleq \text{Adv}$ is compatible in this experiment because $ct \in \mathcal{C}_n \subseteq \mathcal{C}'_n$.

Since ε is nonnegligible, there is $c > 0$ such that $\varepsilon(n) \geq \frac{1}{n^c}$ for infinitely many $n \in \mathbb{N}$. Now notice that for each $n \in \mathbb{N}$ and public key pk , distributions $\text{Enc}_{pk}(\cdot)$ and $\text{Enc}'_{pk}(\cdot)$ agree on every $m \in \mathcal{M}_n$ but m_n^* . Thus, in the experiment $\text{PKE}^{\text{Rand-Same-or-not}}$ for (KeyGen, Enc, Dec), conditioning $m \neq m_n^* \wedge m' \neq m_n^*$, the distribution of ct , which **Adv** takes as input, is identical to that of ct in the same experiment for (KeyGen, Enc', Dec') conditioned on $m \neq m_n^* \wedge m' \neq m_n^*$. So we know

$$\Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [R \text{ wins} \mid m \neq m_n^* \wedge m' \neq m_n^*] = \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [\text{Adv wins} \mid m \neq m_n^* \wedge m' \neq m_n^*]. \quad (3.8)$$

This gives rise to another question: what is the win rate of **Adv** conditioned on $m \neq m_n^* \wedge m' \neq m_n^*$? In fact, since we have (3.7) (which is the “average-case” win rate), we may argue that, conditioning $m \neq m_n^* \wedge m' \neq m_n^*$, the win rate of **Adv** admits a $\frac{1}{2} + \text{nonnegl}(n)$ lower bound, as proven below.

Observation 3.3 (a tight lower bound). For all $n \in \mathbb{N}$,

$$\Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [\text{Adv wins} \mid m \neq m_n^* \wedge m' \neq m_n^*] \geq \left(\frac{1}{2} + \varepsilon(n) - \frac{1}{2^{n-1}} + \frac{1}{2^{2n}} \right) \left(1 + \frac{1}{2^n - 1} \right)^2.$$

Proof. Suppose to the contrary that

$$\Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [\text{Adv wins} \mid m \neq m_n^* \wedge m' \neq m_n^*] < \left(\frac{1}{2} + \varepsilon(n) - \frac{1}{2^{n-1}} + \frac{1}{2^{2n}} \right) \left(1 + \frac{1}{2^n - 1} \right)^2.$$

Thus, by (3.7), we deduce that

$$\begin{aligned} \frac{1}{2} + \varepsilon(n) &\leq \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [\text{Adv wins}] \\ &= \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [m = m_n^* \vee m' = m_n^*] \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [\text{Adv wins} \mid m = m_n^* \vee m' = m_n^*] \\ &\quad + \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [m \neq m_n^* \wedge m' \neq m_n^*] \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [\text{Adv wins} \mid m \neq m_n^* \wedge m' \neq m_n^*] \\ &< \left(1 - \left(1 - \frac{1}{2^n} \right)^2 \right) \cdot 1 + \left(1 - \frac{1}{2^n} \right)^2 \cdot \left(\frac{1}{2} + \varepsilon(n) - \frac{1}{2^{n-1}} + \frac{1}{2^{2n}} \right) \left(1 + \frac{1}{2^n - 1} \right)^2 \\ &= \frac{1}{2} + \varepsilon(n), \end{aligned}$$

which is absurd. ■

cont'd. Following [Observation 3.3](#), we may find a prettier lower bound for the win rate of **Adv** conditioning $m \neq m_n^* \wedge m' \neq m_n^*$. For infinitely many $n \in \mathbb{N}$, we have

$$\begin{aligned}
 \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [\mathbf{Adv} \text{ wins} \mid m \neq m_n^* \wedge m' \neq m_n^*] &\geq \left(\frac{1}{2} + \varepsilon(n) - \frac{1}{2^{n-1}} + \frac{1}{2^{2n}} \right) \left(1 + \frac{1}{2^n - 1} \right)^2 \\
 &> \frac{1}{2} + \frac{1}{n^c} - \frac{1}{2^{n-1}} \\
 &= \frac{1}{2} + \frac{1}{2n^c} + \left(\frac{1}{2n^c} - \frac{1}{2^{n-1}} \right) \\
 &> \frac{1}{2} + \frac{1}{2n^c}. \quad (\because \frac{1}{2n^c} > \frac{1}{2^{n-1}} \text{ for sufficiently large } n \in \mathbb{N})
 \end{aligned} \tag{3.9}$$

Putting all together, we see that for infinitely many $n \in \mathbb{N}$, the win rate of R is hence bounded below by

$$\begin{aligned}
 &\Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [R \text{ wins}] \\
 &\geq \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [R \text{ wins} \wedge m \neq m_n^* \wedge m' \neq m_n^*] \\
 &= \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [m \neq m_n^* \wedge m' \neq m_n^*] \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [R \text{ wins} \mid m \neq m_n^* \wedge m' \neq m_n^*] \\
 &= \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [m \neq m_n^* \wedge m' \neq m_n^*] \Pr_{m, m' \xleftarrow{\$} \mathcal{M}_n} [\mathbf{Adv} \text{ wins} \mid m \neq m_n^* \wedge m' \neq m_n^*] \quad (\text{by (3.8)}) \\
 &\geq \left(1 - \frac{1}{2^n} \right)^2 \left(\frac{1}{2} + \frac{1}{2n^c} \right) \quad (\text{by the assumption } |\mathcal{M}_n| = 2^n \text{ and (3.9)}) \\
 &> \left(1 - \frac{1}{2^{n-1}} \right) \left(\frac{1}{2} + \frac{1}{2n^c} \right) \\
 &= \frac{1}{2} + \frac{1}{2n^c} - \frac{1}{2^n} - \frac{1}{n^c 2^n} \\
 &\geq \frac{1}{2} + \frac{1}{2n^c} - \frac{1}{2^n} - \frac{1}{2^n} \quad (\because n \geq 1) \\
 &= \frac{1}{2} + \frac{1}{4n^c} + \left(\frac{1}{4n^c} - \frac{1}{2^{n-1}} \right) \\
 &> \frac{1}{2} + \frac{1}{4n^c} \quad (\because \frac{1}{4n^c} > \frac{1}{2^{n-1}} \text{ for sufficiently large } n \in \mathbb{N}) \\
 &= \frac{1}{2} + \Omega(n^{-c}),
 \end{aligned}$$

which means that R has nonnegligible advantage in $\text{PKE}^{\text{Rand-Same-or-not}}$.

By definition, this means that $(\text{KeyGen}, \text{Enc}, \text{Dec})$ is not Rand-Same-or-not-secure, which contradicts our assumption. ■

Claim 3.4. $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ is not IND-CPA-secure.

Proof. We then construct a PPT adversary **Adv** that breaks the IND-CPA security of $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ with nonnegligible advantage, as illustrated in the following experiment. ■

cont'd.

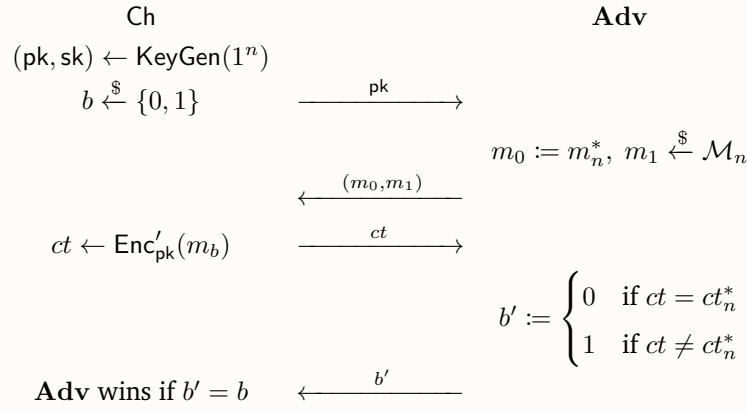


Figure 6: Experiment $\text{PKE}^{IND-CPA}$

Clearly **Adv** is a PPT algorithm. Also, it's straightforward that in this experiment, for all $n \in \mathbb{N}$,

$$\begin{aligned}
 & \Pr_{\substack{m_1 \xleftarrow{\$} \mathcal{M}_n \\ b \xleftarrow{\$} \{0,1\}}} [\mathbf{Adv} \text{ wins}] \\
 & \geq \Pr_{\substack{m_1 \xleftarrow{\$} \mathcal{M}_n \\ b \xleftarrow{\$} \{0,1\}}} [\mathbf{Adv} \text{ wins} \wedge m_1 \neq m_n^*] \\
 & = \Pr_{m_1 \xleftarrow{\$} \mathcal{M}_n} [m_1 \neq m_n^*] \Pr_{\substack{m_1 \xleftarrow{\$} \mathcal{M}_n \\ b \xleftarrow{\$} \{0,1\}}} [\mathbf{Adv} \text{ wins} \mid m_1 \neq m_n^*] \\
 & = \left(1 - \frac{1}{2^n}\right) \left(\frac{1}{2} \Pr_{\substack{m_1 \xleftarrow{\$} \mathcal{M}_n \\ b \xleftarrow{\$} \{0,1\}}} [\mathbf{Adv} \text{ wins} \mid m_1 \neq m_n^* \wedge b = 0] + \frac{1}{2} \Pr_{\substack{m_1 \xleftarrow{\$} \mathcal{M}_n \\ b \xleftarrow{\$} \{0,1\}}} [\mathbf{Adv} \text{ wins} \mid m_1 \neq m_n^* \wedge b = 1] \right) \\
 & = \left(1 - \frac{1}{2^n}\right) \left(\frac{1}{2} \Pr_{(pk, sk) \leftarrow \text{KeyGen}(1^n)} [\underbrace{\text{Enc}'_{pk}(m_n^*) = ct_n^*}_{\equiv \top \text{ (by def. of Enc')}}] + \frac{1}{2} \Pr_{\substack{m_1 \xleftarrow{\$} \mathcal{M}_n \\ (pk, sk) \leftarrow \text{KeyGen}(1^n)}} [\underbrace{\text{Enc}'_{pk}(m_1) \neq ct_n^* \mid m_1 \neq m_n^*}_{\equiv \top \text{ (by def. of Enc')}}] \right) \\
 & = \frac{1}{2} + \left(\frac{1}{2} - \frac{1}{2^n}\right),
 \end{aligned}$$

where $\frac{1}{2} - \frac{1}{2^n}$ is clearly nonnegligible. This means that **Adv** wins in $\text{PKE}^{IND-CPA}$ with nonnegligible advantage. Therefore, by definition, $(\text{KeyGen}, \text{Enc}', \text{Dec}')$ not IND-CPA-secure. ■

Hence we are done.

⊗

Problem 4 (CPA-secure Symmetric-Key Encryption (SKE) from PRF).

Consider the PRF-based SKE scheme constructed from a PRF $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$, defined as following : (general key length/output length also works, but we here just consider the simple case.)

- $\text{KeyGen}(1^n)$: sample a uniformly random $k \in \{0, 1\}^n$.
- $\text{Enc}_k(m)$: choose a uniformly random $r \in \{0, 1\}^n$, and output $ct = (r, F_k(r) \oplus m)$.
- $\text{Dec}_k((r, t))$: output $F_k(r) \oplus t$.

(i) Show that this scheme satisfies the correctness condition.

(ii) Show that this scheme is CPA-secure.

Answer (i). This is extremely trivial. To see this, notice that for all $n \in \mathbb{N}$ and $m \in \mathcal{M}_n$,

$$\begin{aligned} \Pr_{k \leftarrow \text{KeyGen}(1^n)} [\text{Dec}_k(\text{Enc}_k(m)) = m] &= \Pr_{k \leftarrow \text{KeyGen}(1^n)} [\text{Dec}_k((r, F_k(r) \oplus m)) = m] \\ &= \Pr_{k \leftarrow \text{KeyGen}(1^n)} [F_k(r) \oplus (F_k(r) \oplus m) = m] \\ &= \Pr_{k \leftarrow \text{KeyGen}(1^n)} [m = m] \\ &= 1 \geq 1 - 0. \end{aligned}$$

Since 0 is a negligible function of n , the SKE scheme $(\text{KeyGen}, \text{Enc}, \text{Dec})$ is hence correct. $\circledast$

Answer (ii). We prove by contradiction. Suppose to the contrary that the SKE scheme $(\text{KeyGen}, \text{Enc}, \text{Dec})$ is not CPA-secure. Then by definition, there exist a PPT adversary Adv and a nonnegligible function ε such that for every $n \in \mathbb{N}$, in the experiment SKE^{CPA} we have

$$\Pr[\text{Adv wins}] = \frac{1}{2} + \varepsilon(n). \quad (4.1)$$

We then construct a PPT oracle distinguisher (a reduction) $\mathcal{R}$ that breaks the pseudorandomness of the function F with nonnegligible advantage, as illustrated in the following PRF game G^{PRF} .

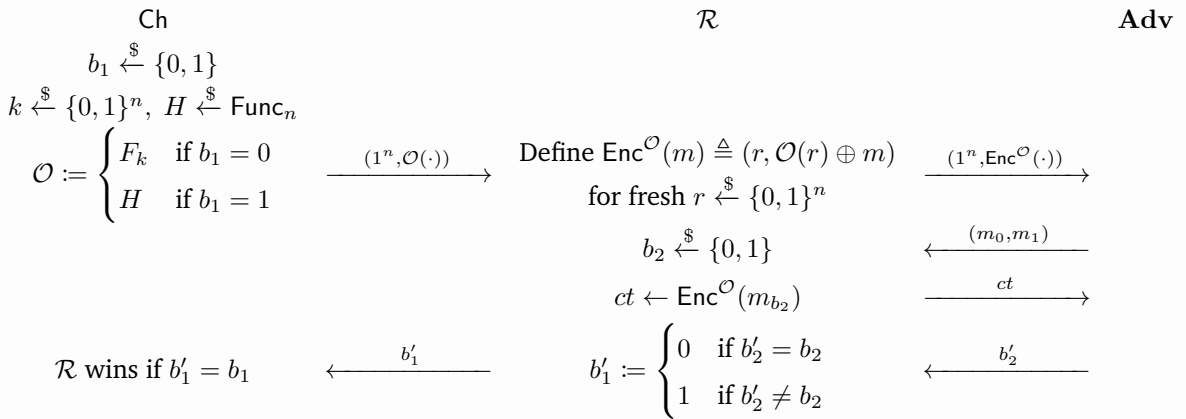


Figure 7: Game G^{PRF}

Note that Func_n denotes the set of all functions mapping n -bit strings to n -bit strings.

Clearly $\mathcal{R}$ is a PPT algorithm since Adv is a PPT algorithm. To see $\mathcal{R}$ wins G^{PRF} with nonnegligible advantage, observe that for all $n \in \mathbb{N}$:

- If $b_1 = 0$ and thus $\mathcal{O} := F_k$ for uniform $k \xleftarrow{\$} \{0, 1\}^n$, then

$$\text{Enc}^{\mathcal{O}} = m \mapsto (U_n, F_k(U_n) \oplus m) = \text{Enc}_k.$$

This means that the encryption oracle access $\text{Enc}^{\mathcal{O}}(\cdot)$ and ciphertext $ct \leftarrow \text{Enc}^{\mathcal{O}}(m_{b_2})$ that are given by $\mathcal{R}$ to Adv are distributed identically to those in the experiment SKE^{CPA} . That is, the view of Adv in G^{PRF} is distributed as if Adv were interacting with the challenger in the experiment SKE^{CPA} . Consequently, $\mathcal{R}$ wins here iff Adv correctly guesses b_2 , whose probability is equal to the probability that Adv wins in SKE^{CPA} , that is,

$$\Pr_{b_1 \xleftarrow{\$} \{0,1\}} [\mathcal{R} \text{ wins} \mid b_1 = 0] = \Pr_{b_1, b_2 \xleftarrow{\$} \{0,1\}} [b'_2 = b_2 \mid b_1 = 0] = \Pr_{b_1, b_2 \xleftarrow{\$} \{0,1\}} [\text{Adv wins in } \text{SKE}^{CPA}] \stackrel{(4.1)}{=} \frac{1}{2} + \varepsilon(n). \quad (4.2)$$

- If $b_1 = 1$ and thus $\mathcal{O} := H$ for H chosen uniformly from Func_n , then $\text{Enc}^{\mathcal{O}} = m \mapsto (U_n, H(U_n) \oplus m)$, and the ciphertext $ct = (r, H(r) \oplus m_{b_2})$ where $r \xleftarrow{\$} \{0, 1\}^n$. For any fresh $r' \xleftarrow{\$} \{0, 1\}^n$, since H is uniformly chosen from Func_n , $H(r') \in \{0, 1\}^n$ is uniformly random and independent of r' . Therefore, even though Adv has access to $\text{Enc}^{\mathcal{O}} = m \mapsto (U_n, H(U_n) \oplus m)$ and can thus infer the value of $H(r'')$ (for a fresh $r'' \xleftarrow{\$} \{0, 1\}^n$) for each query to $\text{Enc}^{\mathcal{O}}$, $H(r)$ (which is used to pad m_{b_2}) is still independent of the message pair (m_0, m_1) chosen by Adv . Lastly, since $H(r)$ is uniform (as reasoned above) and independent of m_{b_2} ($\because H(r) \perp\!\!\!\perp (m_0, m_1)$), by the uniformity property proven in Problem 1-(ii) in HW1, $H(r) \oplus m_{b_2}$ is uniformly random and, more importantly, independent of m_{b_2} and thus b_2 . Hence, the ciphertext $ct = (r, H(r) \oplus m_{b_2})$ given by $\mathcal{R}$ to Adv is independent of b_2 , and so does the output b'_2 of Adv . Since $b_2 \xleftarrow{\$} \{0, 1\}$ is uniformly random, this implies that $\Pr[b'_2 = \overline{b_2} \mid b_1 = 1] = \frac{1}{2}$. Intuitively speaking, ct leaks to Adv zero information of the bit b_2 , so Adv correctly guesses the value of b_2 with probability only $\frac{1}{2}$. In other words,

$$\begin{aligned} \Pr_{b_1 \xleftarrow{\$} \{0,1\}} [\mathcal{R} \text{ wins} \mid b_1 = 1] &= \Pr_{b_1, b_2 \xleftarrow{\$} \{0,1\}} [b'_2 = \overline{b_2} \mid b_1 = 1] \\ &= \Pr_{\substack{b_2 \xleftarrow{\$} \{0,1\} \\ H \xleftarrow{\$} \text{Func}_n \\ r \xleftarrow{\$} \{0,1\}^n \\ (m_0, m_1) \leftarrow \text{Adv}(1^n, \text{Enc}^H(\cdot))}} [\text{Adv}(\text{Enc}^H(\cdot), (r, H(r) \oplus m_{b_2})) = \overline{b_2}] = \frac{1}{2} \end{aligned} \quad (4.3)$$

since $b_2 \in \{0, 1\}$ is independent of LHS and is uniformly random.

Putting both together, we see that for all $n \in \mathbb{N}$

$$\Pr_{b_1 \xleftarrow{\$} \{0,1\}} [\mathcal{R} \text{ wins}] = \frac{1}{2} \Pr_{b_1 \xleftarrow{\$} \{0,1\}} [\mathcal{R} \text{ wins} \mid b_1 = 0] + \frac{1}{2} \Pr_{b_1 \xleftarrow{\$} \{0,1\}} [\mathcal{R} \text{ wins} \mid b_1 = 1] \stackrel{(4.2)+(4.3)}{=} \frac{1}{2} \left(\frac{1}{2} + \varepsilon(n) \right) + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2} + \frac{\varepsilon(n)}{2},$$

where the advantage $\frac{\varepsilon(n)}{2}$ of $\mathcal{R}$ in G^{PRF} is nonnegligible. Therefore, by definition, F is not a PRF, contradicting our assumption. $\otimes$

Problem 5 (CCA-security).

- (i) Prove that El Gamal PKE is not IND-CCA-secure.
- (ii) Prove that Regev's PKE is not IND-CCA-secure.

Answer (i).

Intuition. El Gamal PKE is not IND-CCA-secure since, given a ciphertext, the adversary can forge another ciphertext that decrypts to some plaintext, from which the adversary can then recover the original plaintext.

Let $(\text{KeyGen}, \text{Enc}, \text{Dec})$ denote the EL Gamal PKE scheme with an ensemble $\mathcal{G} = \{(\mathbb{G}_n, q_n, g_n)\}_{n \in \mathbb{N}}$ of group parameters, where $\mathbb{G}_n$ is a cyclic group of prime order $q_n = O(2^n)$ and g_n is a generator of $\mathbb{G}_n$. Denote also the identity of the cyclic group $\mathbb{G}_n$ by e_n . Note that $g_n \neq e_n$ for all $n \in \mathbb{N}$.

To prove the problem, we construct a PPT adversary Adv that breaks the IND-CCA security of $(\text{KeyGen}, \text{Enc}, \text{Dec})$ with nonnegligible advantage, as illustrated in the following experiment $\text{PKE}^{\text{IND-CCA}}$.

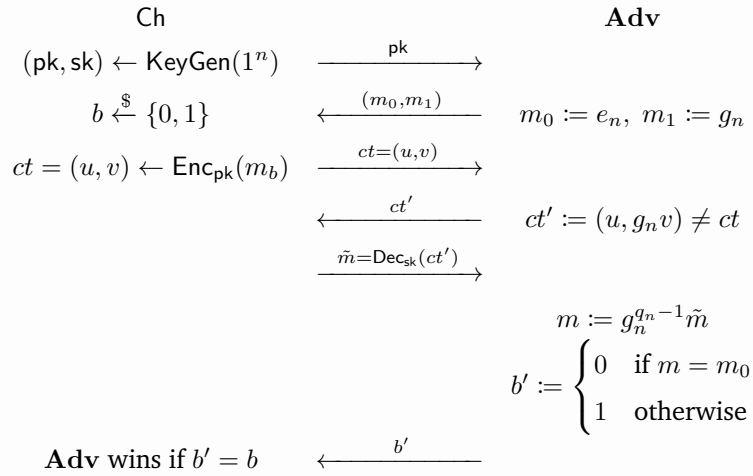


Figure 8: Experiment $\text{PKE}^{\text{IND-CCA}}$

Note that in the experiment above, it always holds that $m_0 \neq m_1$ and $ct \neq ct'$ since otherwise we would have $g_n = e_n$.

The only potentially time-consuming operation in Adv is the computation of $g_n^{q_n-1}$, which can be computed by fast exponentiation (aka exponentiation by squaring) in $O(\log q_n) = O(\log(O(2^n))) = O(n)$ time. Therefore, it's clear that Adv is a PPT algorithm.

As for the advantage of Adv , observe that for all $n \in \mathbb{N}$

$$\begin{aligned}
 m &= g_n^{q_n-1} \tilde{m} = g_n^{q_n-1} \text{Dec}_{sk}((u, g_n v)) \\
 &= g_n^{q_n-1} \text{Dec}_{sk}((g_n^r, g_n (g^{sk})^r m_b)) \\
 &= g_n^{q_n-1} (g_n (g_n^{sk})^r m_b) (g_n^r)^{-sk} \\
 &= g_n^{q_n} m_b && (\because \mathbb{G}_n \text{ is cyclic} \implies \mathbb{G}_n \text{ is abelian}) \\
 &= m_b, && (\because g_n^{q_n} = e_n)
 \end{aligned} \tag{5.1}$$

where $r \xleftarrow{\$} \mathbb{Z}_{q_n}$ is fresh. Besides, since $m_0 \neq m_1$, $m_{b_1} = m_{b_2}$ iff $b_1 = b_2$ for all $b_1, b_2 \in \{0, 1\}$. Therefore,

$$m = m_0 \stackrel{(5.1)}{\iff} m_b = m_0 \iff b = 0,$$

so we have

$$b' = [\neg(m = m_0)] = [\neg(b = 0)] = [b = 1] = b. \quad (5.2)$$

For all $n \in \mathbb{N}$, the win rate of **Adv** is hence

$$\Pr[\mathbf{Adv} \text{ wins}] = \Pr_{b \xleftarrow{\$} \{0,1\}} [b' = b] \stackrel{\text{by (5.2)}}{=} \Pr_{b \xleftarrow{\$} \{0,1\}} [\top] = 1 = \frac{1}{2} + \frac{1}{2},$$

where the advantage $\frac{1}{2}$ of **Adv** is a nonnegligible function of n .

By definition, this means that (KeyGen, Enc, Dec) is not IND-CCA secure. ⊛

Answer (ii).

Intuition. Regev's PKE is not IND-CCA-secure since, given a ciphertext, the adversary can forge another ciphertext that decrypts to some plaintext, from which the adversary can then recover the original plaintext with high probability.

Let (KeyGen, Enc, Dec) denote Regev's PKE scheme with modulus q_n , $m_n = \Omega(n \log n)$, and discrete Gaussian distribution $\chi \triangleq \Psi_\alpha$ over $\mathbb{Z}_{q_n}$ with standard deviation $\alpha q_n > 2\sqrt{n}$ as the error distribution, for each security parameter $n \in \mathbb{N}$. Also, we further suppose that $m_n = O(\text{poly}(n))$ and $\frac{q_n}{4m_n} \geq 1$. For convenience, denote $q \triangleq q_n$ and $m \triangleq m_n$.

For each $x \in \mathbb{Z}_q$, let $[x]_q$ denote the *central representative* (aka *centered mod*) of x , that is, $[x]_q \in [-\frac{q}{2}, \frac{q}{2})$ is such that $[x]_q \bmod q = x \in \mathbb{Z}_q$. Note that in Regev's PKE scheme, Dec in fact utilizes $[\cdot]_q$ to check whether the unpadded result mod q is closer to 0 or $\lfloor \frac{q}{2} \rfloor$:

$$\text{Dec}_{\text{sk}}(u, v) \triangleq \begin{cases} 0 & \text{if } |[v - usk]_q| < \frac{q}{4} \\ 1 & \text{otherwise} \end{cases}. \quad (5.3)$$

(Intuitively speaking, $|[x]_q|$ denotes the minimal absolute value of the integers in its equivalence class.) To guarantee correctness of Regev's PKE scheme, as shown in [the lecture note by Vinod Vaikuntanathan](#), it's required that

$$\Pr_{e \leftarrow \chi} \left[|[e]_q| < \frac{q}{4m} \right] \geq 1 - \varepsilon(n) \quad (5.4)$$

for some negligible ε . Taking a step further,

$$\Pr_{e \leftarrow \chi^m} \left[\bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] \stackrel{\text{union bound}}{\geq} 1 - \sum_{i=1}^m \left(1 - \Pr_{e_i \leftarrow \chi} \left[|[e_i]_q| < \frac{q}{4m} \right] \right) \stackrel{(5.4)}{\geq} 1 - \sum_{i=1}^m \varepsilon(n) = 1 - m\varepsilon(n). \quad (5.5)$$

Since $m = O(\text{poly}(n))$ and $\varepsilon(n)$ is negligible, $m\varepsilon(n)$ is also negligible. Sloppily speaking, this means that all components of $e \leftarrow \chi^m$ stay within the bound $\frac{q}{4m}$ with $1 - \text{negl}(n)$ probability.

To prove the problem itself, we construct a PPT adversary **Adv** that breaks the IND-CCA security of (KeyGen, Enc, Dec) with nonnegligible advantage, as illustrated in the following experiment

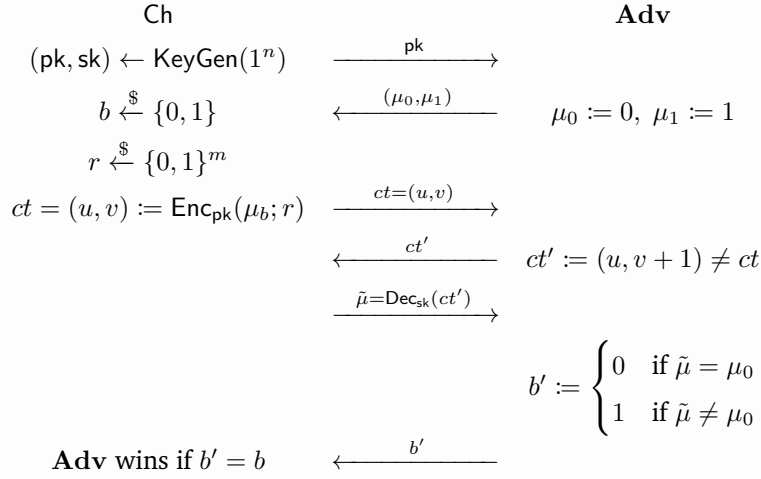


Figure 9: Experiment PKE^{IND-CCA}

Note that in the experiment above, it always holds that $\mu_0 \neq \mu_1$ and $ct \neq ct'$ since otherwise we would have $0 = 1 \in \mathbb{Z}_q$.

Clearly **Adv** is a PPT algorithm. Observe that in the experiment, if it happens that ct' decrypts to the same message μ_b (i.e. $\tilde{\mu} = \mu_b$), then $b' = b$ (note that $\tilde{\mu} = \mu_1 \implies \tilde{\mu} \neq \mu_0$ as $\mu_0 \neq \mu_1$), leading to a win for **Adv**. To see that $\tilde{\mu} = \mu_b$ happens with high probability, we make the following observation.

Claim 5.1. $\Pr[\tilde{\mu} = \mu_b] \geq 1 - \text{negl}(n)$.

Proof. By construction, $pk = (A, As + e)$ and $sk = s$, where $A \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$, $s \xleftarrow{\$} \mathbb{Z}_q^n$, and $e \leftarrow \chi^m$. Also, $(u, v) = \text{Enc}_{pk}(\mu_b; r) = (r^T A, r^T (As + e) + \mu_b \lfloor \frac{q}{2} \rfloor)$, where $r \xleftarrow{\$} \{0, 1\}^m$.

Compute that

$$\begin{aligned}
 \tilde{\mu} = \text{Dec}_{sk}(u, v + 1) &= \text{Dec}_s \left(r^T A, r^T (As + e) + \mu_b \lfloor \frac{q}{2} \rfloor + 1 \right) \\
 &\stackrel{(5.3)}{=} \begin{cases} 0 & \text{if } \left| (r^T (As + e) + \mu_b \lfloor \frac{q}{2} \rfloor + 1) - (r^T A)s \right|_q < \frac{q}{4} \\ 1 & \text{otherwise} \end{cases} \\
 &= \begin{cases} 0 & \text{if } \left| \left[\mu_b \lfloor \frac{q}{2} \rfloor + r^T e + 1 \right]_q \right| < \frac{q}{4} \\ 1 & \text{otherwise} \end{cases}. \tag{5.6}
 \end{aligned}$$

We analyze the probability that $\left| \left[\mu_b \lfloor \frac{q}{2} \rfloor + r^T e + 1 \right]_q \right| < \frac{q}{4}$ by considering the two cases $\mu_b = 0$ and $\mu_b = 1$.

Observation 5.2. Conditioning on $r \neq 1^m$ and $|[e_i]_q| < \frac{q}{4m}$ for every $i \in [m]$, we have

$$\left| \left[\mu_b \lfloor \frac{q}{2} \rfloor + r^T e + 1 \right]_q \right| \begin{cases} < \frac{q}{4} & \text{if } \mu_b = 0 \\ \geq \frac{q}{4} & \text{if } \mu_b = 1 \end{cases}.$$

■

cont'd.

Proof of Observation 5.2. First and foremost, we treat all numbers in $\mathbb{Z}_q$ as integers in $\mathbb{Z}$, allowing ourselves to find an interval bound for $\left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1\right]_q$ separately in the two cases $\mu_b = 0$ and $\mu_b = 1$.

As $r \neq 1^m$, we have $r_j = 0$ for some $j \in [m]$. Thus,

$$\begin{aligned} \left| \left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1 \right]_q \right| &= \left| \left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1}^m r_i e_i + 1 \right]_q \right| \\ &= \left| \left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1, i \neq j}^m r_i e_i + 1 \right]_q \right| = \left| \left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \right]_q \right|. \end{aligned} \quad (5.7)$$

Then, since $r_i \in \{0, 1\}$ and $-\frac{q}{4m} < [e_i]_q < \frac{q}{4m}$ for every $i \in [m]$,

$$-\frac{q}{4} < -\frac{q}{4} + \frac{q}{4m} + 1 = \sum_{i=1, i \neq j}^m 1 \cdot \left(-\frac{q}{4m}\right) + 1 < \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 < \sum_{i=1, i \neq j}^m 1 \cdot \frac{q}{4m} + 1 = \frac{q}{4} - \frac{q}{4m} + 1 \stackrel{\because \frac{q}{4m} \geq 1}{\leq} \frac{q}{4},$$

which further implies that

$$-\left\lceil \frac{q}{4} - 1 \right\rceil \leq \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \leq \left\lceil \frac{q}{4} - 1 \right\rceil \quad (5.8)$$

as $\sum_{i=1, i \neq j}^m r_i [e_i]_q + 1$ is an integer.

We now consider the two cases separately.

- If $\mu_b = 0$, it's immediate that

$$\begin{aligned} \left| \left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \right]_q \right| &= \left| \left[\sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \right]_q \right| \\ &= \left| \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \right| \quad (\because -\frac{q}{2} \leq \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 < \frac{q}{2} \text{ (by (5.8))}) \\ &\leq \left\lceil \frac{q}{4} - 1 \right\rceil \quad (\text{by (5.8) again}) < \frac{q}{4}. \end{aligned}$$

- If $\mu_b = 1$, then by (5.8) we have

$$\left\lfloor \frac{q}{2} \right\rfloor - \left\lceil \frac{q}{4} - 1 \right\rceil \leq \left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \leq \left\lfloor \frac{q}{2} \right\rfloor + \left\lceil \frac{q}{4} - 1 \right\rceil.$$

Taking centered mod $[\cdot]_q$, we get

$$\begin{aligned} \left[\left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \right]_q &\leq \left\lfloor \frac{q}{2} \right\rfloor + \left\lceil \frac{q}{4} - 1 \right\rceil - q < \frac{q}{2} + \left(\frac{q}{4} - 1 + 1 \right) - q = -\frac{q}{4} \\ \vee \left[\left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \right]_q &\geq \left\lfloor \frac{q}{2} \right\rfloor - \left\lceil \frac{q}{4} - 1 \right\rceil > \left(\frac{q}{2} - 1 \right) - \left(\frac{q}{4} - 1 \right) = \frac{q}{4}. \end{aligned}$$

Hence,

$$\left| \left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1 \right]_q \right| \stackrel{(5.7)}{=} \left| \left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \right]_q \right| = \left| \left[\left\lfloor \frac{q}{2} \right\rfloor + \sum_{i=1, i \neq j}^m r_i [e_i]_q + 1 \right]_q \right| \geq \frac{q}{4}.$$

cont'd.

What about the probability that each condition holds? In fact,

$$\Pr_{r \leftarrow \mathbb{S}_{\{0,1\}^m}} [r \neq 1^m] = 1 - \frac{1}{2^m} = 1 - \frac{1}{2^{\Omega(n \log n)}} = 1 - \frac{1}{2^{\omega(1) \log n}} = 1 - \frac{1}{n^{\omega(1)}} = 1 - \text{negl}(n). \quad (5.9)$$

Also, (5.5) already gives us that

$$\Pr_{e \leftarrow \chi^m} \left[\bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] = 1 - \text{negl}(n). \quad (5.10)$$

Hence, the probability that $\tilde{\mu} = \mu_b$ is

$$\begin{aligned} & \Pr[\tilde{\mu} = \mu_b] \\ &= \Pr[\mu_b = 0] \Pr[\tilde{\mu} = 0] + \Pr[\mu_b = 1] \Pr[\tilde{\mu} = 1] \\ &= \Pr[\mu_b = 0] \Pr \left[\left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1 \right]_q < \frac{q}{4} \right] + \Pr[\mu_b = 1] \Pr \left[\left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1 \right]_q \geq \frac{q}{4} \right] \quad (\text{by (5.6)}) \\ &\geq \Pr[\mu_b = 0] \Pr \left[\left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1 \right]_q < \frac{q}{4} \wedge r \neq 1^m \wedge \bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] \\ &\quad + \Pr[\mu_b = 1] \Pr \left[\left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1 \right]_q \geq \frac{q}{4} \wedge r \neq 1^m \wedge \bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] \\ &= \Pr[\mu_b = 0] \Pr_{r \leftarrow \mathbb{S}_{\{0,1\}^m}} [r \neq 1^m] \Pr_{e \leftarrow \chi^m} \left[\bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] \Pr \left[\underbrace{\left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1 \right]_q < \frac{q}{4}}_{\equiv \top \text{ (by Observation 5.2)}} \mid r \neq 1^m \wedge \bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] \\ &\quad + \Pr[\mu_b = 1] \Pr_{r \leftarrow \mathbb{S}_{\{0,1\}^m}} [r \neq 1^m] \Pr_{e \leftarrow \chi^m} \left[\bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] \Pr \left[\underbrace{\left[\mu_b \left\lfloor \frac{q}{2} \right\rfloor + r^T e + 1 \right]_q \geq \frac{q}{4}}_{\equiv \top \text{ (by Observation 5.2)}} \mid r \neq 1^m \wedge \bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] \\ &= \Pr_{r \leftarrow \mathbb{S}_{\{0,1\}^m}} [r \neq 1^m] \Pr_{e \leftarrow \chi^m} \left[\bigwedge_{i=1}^m |[e_i]_q| < \frac{q}{4m} \right] \\ &= (1 - \text{negl}(n))(1 - \text{negl}(n)) \quad (\text{by (5.9) and (5.10)}) \\ &\geq 1 - \text{negl}(n) - \text{negl}(n) \\ &= 1 - \text{negl}(n), \end{aligned}$$

as claimed. ■

FINALLY, as observed above, $\tilde{\mu} = \mu_b$ implies **Adv** wins in the experiment $\text{PKE}^{\text{IND-CCA}}$, so by [Claim 5.1](#) we deduce that

$$\Pr[\mathbf{Adv} \text{ wins}] \geq \Pr[\tilde{\mu} = \mu_b] \stackrel{\text{by Claim 5.1}}{\geq} 1 - \text{negl}(n) = \frac{1}{2} + \left(\frac{1}{2} - \text{negl}(n) \right).$$

Since $\frac{1}{2} - \text{negl}(n)$ is clearly nonnegligible, **Adv**, a PPT algorithm, wins with nonnegligible advantage in the experiment $\text{PKE}^{\text{IND-CCA}}$. Hence, by definition, Regev's PKE scheme (KeyGen, Enc, Dec) is not IND-CCA secure. ⊛

Problem 6 (Complete the proof in the class).

- (i) Recall that in the proof of Regev's PKE scheme being IND-CPA secure we first deduced from DLWE assumption that

$$\left(A, b_{\text{pk}}, M_0, M_1, r^T A, r^T b_{\text{pk}} + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) \approx_c \left(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right). \quad (6.1)$$

Formally prove the equation (6.1) by constructing a reduction.

- (ii) Then we applied the Leftover Hash Lemma to show that

$$\left(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) \approx_s (A, u, M_0, M_1, u_1, u_2). \quad (6.2)$$

Formally prove the equation (6.2) by constructing a reduction.

Answer (i). Assume $M_0, M_1 \in \{0, 1\}$ are fixed. Recall that $A \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$, $b_{\text{pk}} = As + e$, $s \xleftarrow{\$} \mathbb{Z}_q^n$, $e \leftarrow \chi^m$, $u \xleftarrow{\$} \mathbb{Z}_q^m$, $r \xleftarrow{\$} \{0, 1\}^m$, and $b \xleftarrow{\$} \{0, 1\}$, where $n \in \mathbb{N}$ is the security parameter, $q \triangleq q_n$ is a modulus, $m \triangleq m_n = \Omega(n \log n)$, and $\chi \triangleq \chi_n$ is an error distribution over $\mathbb{Z}_q$.

We prove (6.1) by contradiction. Suppose to the contrary that these two distributions (they are in fact ensembles) are not **computationally** indistinguishable. By definition, there exists a PPT distinguisher $\mathcal{D}$ such that

$$\varepsilon(n) \triangleq \left| \Pr \left[\mathcal{D} \left(A, b_{\text{pk}}, M_0, M_1, r^T A, r^T b_{\text{pk}} + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] - \Pr \left[\mathcal{D} \left(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] \right|$$

is nonnegligible. Expanding $b_{\text{pk}} = As + e$ to write out all independent random variables, we have

$$\varepsilon(n) = \left| \Pr \left[\mathcal{D} \left(A, As + e, M_0, M_1, r^T A, r^T (As + e) + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] - \Pr \left[\mathcal{D} \left(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] \right|. \quad (6.3)$$

We construct a PPT distinguisher (a reduction) $\mathcal{R}$ that breaks the decisional $\text{LWE}_{n,m,q,\chi}$ (DLWE) assumption for parameters n, m, q, χ (i.e., that distinguishes the ensembles $\{\text{LWE}_{n,m,q,\chi}\}_{n \in \mathbb{N}}$ and $\{U_{\mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m}\}_{n \in \mathbb{N}}$ with nonnegligible advantage), as illustrated in the following experiment.

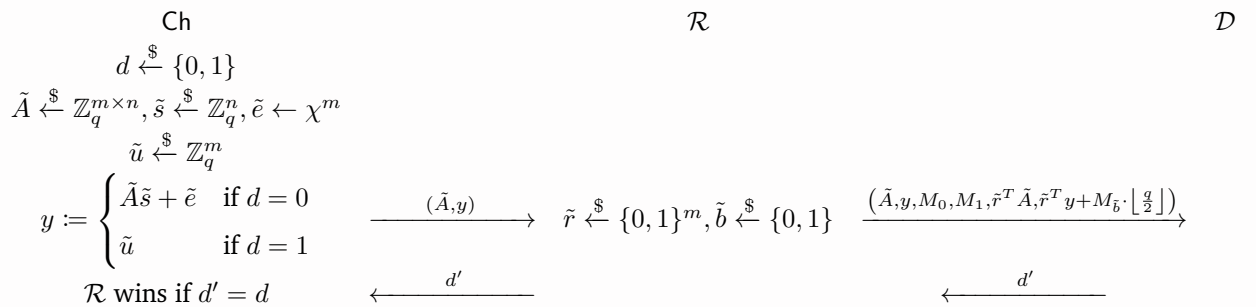


Figure 10: Experiment for decisional $\text{LWE}_{n,m,q,\chi}$ assumption

Clearly $\mathcal{R}$ is a PPT algorithm since $\mathcal{D}$ is a PPT algorithm. Observe that in this experiment, for all $n \in \mathbb{N}$:

- If $d = 0$, then the distribution of

$$\left(\tilde{A}, y, M_0, M_1, \tilde{r}^T \tilde{A}, \tilde{r}^T y + M_{\tilde{b}} \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = \left(\tilde{A}, \tilde{A}\tilde{s} + \tilde{e}, M_0, M_1, \tilde{r}^T \tilde{A}, \tilde{r}^T (\tilde{A}\tilde{s} + \tilde{e}) + M_{\tilde{b}} \cdot \left\lfloor \frac{q}{2} \right\rfloor \right)$$

given by $\mathcal{R}$ to $\mathcal{D}$ is identical to that of $(A, As + e, M_0, M_1, r^T A, r^T (As + e) + M_b \cdot \lfloor \frac{q}{2} \rfloor)$ in (6.3).

- If $d = 1$, then the distribution of

$$\left(\tilde{A}, y, M_0, M_1, \tilde{r}^T \tilde{A}, \tilde{r}^T y + M_{\tilde{b}} \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = \left(\tilde{A}, \tilde{u}, M_0, M_1, \tilde{r}^T \tilde{A}, \tilde{r}^T \tilde{u} + M_{\tilde{b}} \cdot \left\lfloor \frac{q}{2} \right\rfloor \right)$$

given by $\mathcal{R}$ to $\mathcal{D}$ is identical to that of $(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \lfloor \frac{q}{2} \rfloor)$ in (6.3).

Putting both together, we see that for all $n \in \mathbb{N}$, the advantage of $\mathcal{R}$ in distinguishing the ensembles $\{\text{LWE}_{n,m,q,\chi}\}_{n \in \mathbb{N}}$ and $\{U_{\mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m}\}_{n \in \mathbb{N}}$ is hence

$$\begin{aligned} & \left| \Pr[\mathcal{R}(1^n, (\text{LWE}_{n,m,q,\chi})) = 1] - \Pr_{\substack{\tilde{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}, \tilde{u} \xleftarrow{\$} \mathbb{Z}_q^m}} [\mathcal{R}(1^n, (\tilde{A}, \tilde{u})) = 1] \right| \\ &= \left| \Pr_{\substack{\tilde{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}, \tilde{s} \xleftarrow{\$} \mathbb{Z}_q^n, \tilde{e} \xleftarrow{\$} \chi^m}} [\mathcal{R}(1^n, (\tilde{A}, \tilde{A}\tilde{s} + \tilde{e})) = 1] - \Pr_{\substack{\tilde{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}, \tilde{u} \xleftarrow{\$} \mathbb{Z}_q^m}} [\mathcal{R}(1^n, (\tilde{A}, \tilde{u})) = 1] \right| \\ &= \left| \Pr_{\substack{\tilde{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}, \tilde{s} \xleftarrow{\$} \mathbb{Z}_q^n, \tilde{e} \xleftarrow{\$} \chi^m, \tilde{r} \xleftarrow{\$} \{0,1\}^m, \tilde{b} \xleftarrow{\$} \{0,1\}}} \left[\mathcal{D} \left(\tilde{A}, \tilde{A}\tilde{s} + \tilde{e}, M_0, M_1, \tilde{r}^T \tilde{A}, \tilde{r}^T (\tilde{A}\tilde{s} + \tilde{e}) + M_{\tilde{b}} \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] \right. \\ & \quad \left. - \Pr_{\substack{\tilde{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}, \tilde{u} \xleftarrow{\$} \mathbb{Z}_q^m, \tilde{r} \xleftarrow{\$} \{0,1\}^m, \tilde{b} \xleftarrow{\$} \{0,1\}}} \left[\mathcal{D} \left(\tilde{A}, \tilde{u}, M_0, M_1, \tilde{r}^T \tilde{A}, \tilde{r}^T \tilde{u} + M_{\tilde{b}} \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] \right| \\ &= \left| \Pr \left[\mathcal{D} \left(A, As + e, M_0, M_1, r^T A, r^T (As + e) + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] - \Pr \left[\mathcal{D} \left(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] \right| \\ &= \varepsilon(n), \quad (\text{by (6.3)}) \end{aligned}$$

which is nonnegligible. Hence, this contradicts the decisional $\text{LWE}_{n,m,q,\chi}$ assumption, implying (6.1) must be true. $\circledast$

Answer (ii). We prove (6.2) by contradiction. Suppose to the contrary that these two distributions (they are in fact ensembles) are not **statistically** indistinguishable. By definition, there exists a bounded distinguisher $\mathcal{D}$ that runs in some time $t(n)$ such that

$$\mu(n) \triangleq \left| \Pr_{\substack{A \xleftarrow{\$} \mathbb{Z}_q^{m \times n} \\ u \xleftarrow{\$} \mathbb{Z}_q^m \\ r \xleftarrow{\$} \{0,1\}^m \\ b \xleftarrow{\$} \{0,1\}}} \left[\mathcal{D} \left(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = 1 \right] - \Pr_{\substack{A \xleftarrow{\$} \mathbb{Z}_q^{m \times n} \\ u \xleftarrow{\$} \mathbb{Z}_q^m \\ u_1 \xleftarrow{\$} \mathbb{Z}_q^n \\ u_2 \xleftarrow{\$} \mathbb{Z}_q}} [\mathcal{D}(A, u, M_0, M_1, u_1, u_2) = 1] \right| \quad (6.4)$$

is nonnegligible (as a function of n), i.e., $\mathcal{D}$ distinguishes between the two distributions with nonnegligible advantage $\mu(n)$. Additionally, since μ is nonnegligible, there exists $c > 0$ such that $\mu(n) \geq \frac{1}{n^c}$ for infinitely many $n \in \mathbb{N}$, as will be used at the end of our proof.

We then construct a bounded distinguisher (a reduction) $\mathcal{R}$ that, for EACH $n \in \mathbb{N}$, distinguishes $(A^\star, r^T A^\star)$ from uniform $(A^\star, u^\star)$, where $A^\star \xleftarrow{\$} \mathbb{Z}_q^{m \times (n+1)}, r \xleftarrow{\$} \{0,1\}^m, u^\star \xleftarrow{\$} \mathbb{Z}_q^{n+1}$,

with advantage exactly $\mu(n)$, as illustrated in the following distinguishing game G_{n+1}^{IND} .

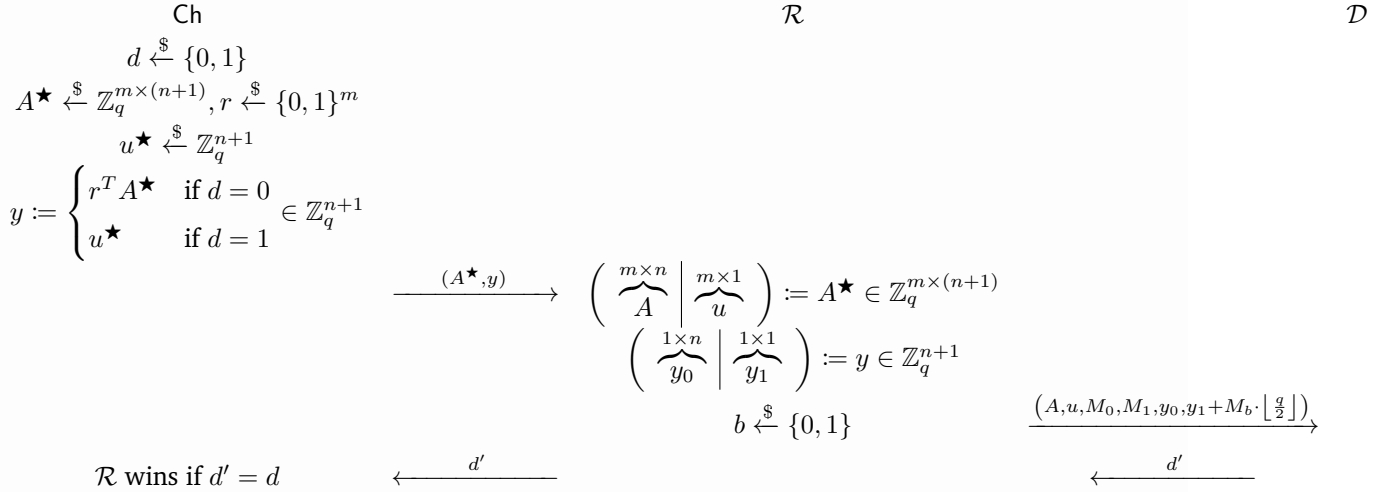


Figure 11: Game G_{n+1}^{IND}

Clearly $\mathcal{R}$ runs in bounded time since $\mathcal{D}$ runs in bounded time. Observe that for EACH $n \in \mathbb{N}$:

- If $d = 0$, observe that

$$\left(\begin{array}{c|c} y_0 & y_1 \end{array} \right) = y = r^T A^\star = r^T \left(\begin{array}{c|c} A & u \end{array} \right) = \left(\begin{array}{c|c} r^T A & r^T u \end{array} \right),$$

which gives $y_0 = r^T A \in \mathbb{Z}_q^n$ and $y_1 = r^T u \in \mathbb{Z}_q$. Clearly $A \in \mathbb{Z}_q^{m \times n}$ and $u \in \mathbb{Z}_q^m$ are uniform and independent of all the other independently sampled random variables in the game (except $A^\star$ of course), so we may redefine $A \xleftarrow{\$} \mathbb{Z}_q^{m \times n}, u \xleftarrow{\$} \mathbb{Z}_q^m$ as fresh random variables and replace $A^\star$ with $\left(\begin{array}{c|c} A & u \end{array} \right)$. Hence, the distribution of

$$\left(A, u, M_0, M_1, y_0, y_1 + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = \left(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right)$$

given by $\mathcal{R}$ to $\mathcal{D}$ is identical to that on the LHS of (6.4).

- If $d = 1$, we split $u^\star = \left(\begin{array}{c|c} u_1 & u_3 \end{array} \right)$, where $u_1 \in \mathbb{Z}_q^n, u_3 \in \mathbb{Z}_q$. Clearly u_1 and u_3 are uniform and independent of all the other independently sampled random variables in the game (except $u^\star$ of course), so we may redefine $u_1 \xleftarrow{\$} \mathbb{Z}_q^n, u_3 \xleftarrow{\$} \mathbb{Z}_q$ as fresh random variables and replace $u^\star$ with $\left(\begin{array}{c|c} u_1 & u_3 \end{array} \right)$. This gives $y_0 = u_1$ and $y_1 = u_3$, so the input of $\mathcal{D}$ becomes

$$\left(A, u, M_0, M_1, y_0, y_1 + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) = \left(A, u, M_0, M_1, u_1, u_3 + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right).$$

Notice that in the last entry, $u_3 \xleftarrow{\$} \mathbb{Z}_q$ is uniformly random and independent of the trailing term $M_b \cdot \lfloor \frac{q}{2} \rfloor$. As $\mathbb{Z}_q$ forms an additive group (in fact, it forms a ring), by the uniformity property proven in Problem 1-(ii) in HW1, their sum $u_3 + M_b \cdot \lfloor \frac{q}{2} \rfloor$ is uniformly random and hence independent of $M_b \cdot \lfloor \frac{q}{2} \rfloor$ (but it may still be dependent of u_3). This allows us to replace the last entry $u_3 + M_b \cdot \lfloor \frac{q}{2} \rfloor$ with a fresh and uniform $u_2 \xleftarrow{\$} \mathbb{Z}_q$, without changing the total distribution of input:

$$\left(A, u, M_0, M_1, u_1, u_3 + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \right) \stackrel{d}{=} \left(A, u, M_0, M_1, u_1, u_2 \right),$$

which is distributed identically to the RHS of (6.4).

Putting both together, we see that for EACH $n \in \mathbb{N}$, the advantage of $\mathcal{R}$ in G^{IND}_{n+1} is exactly

$$\begin{aligned}
 & \left| \Pr_{\substack{A^\star \xleftarrow{\$} \mathbb{Z}_q^{m \times (n+1)} \\ r \xleftarrow{\$} \{0,1\}^m}} [\mathcal{R}(A^\star, r^T A^\star) = 1] - \Pr_{\substack{A^\star \xleftarrow{\$} \mathbb{Z}_q^{m \times (n+1)} \\ u^\star \xleftarrow{\$} \mathbb{Z}_q^{n+1}}} [\mathcal{R}(A^\star, u^\star) = 1] \right| \\
 &= \left| \Pr_{\substack{A \xleftarrow{\$} \mathbb{Z}_q^{m \times n} \\ u \xleftarrow{\$} \mathbb{Z}_q^m \\ r \xleftarrow{\$} \{0,1\}^m \\ b \xleftarrow{\$} \{0,1\}}} \left[\mathcal{D}\left(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \left\lfloor \frac{q}{2} \right\rfloor\right) = 1 \right] - \Pr_{\substack{A \xleftarrow{\$} \mathbb{Z}_q^{m \times n} \\ u \xleftarrow{\$} \mathbb{Z}_q^m \\ u_1 \xleftarrow{\$} \mathbb{Z}_q^n \\ u_2 \xleftarrow{\$} \mathbb{Z}_q}} [\mathcal{D}(A, u, M_0, M_1, u_1, u_2) = 1] \right| \\
 &= \mu(n) \quad (\text{by (6.4)}).
 \end{aligned}$$

Denote by $t'(n)$ the running time of $\mathcal{R}$. For EACH $n \in \mathbb{N}$, the existence of the time $t'(n)$ distinguisher $\mathcal{R}$ with advantage $\mu(n)$ implies

$$(A^\star, r^T A^\star) \not\approx_{t'(n), \mu(n)} (A^\star, u^\star),$$

where $A^\star \xleftarrow{\$} \mathbb{Z}_q^{m \times (n+1)}$, $r \xleftarrow{\$} \{0,1\}^m$, $u^\star \xleftarrow{\$} \mathbb{Z}_q^{n+1}$. By the following lemma (which is proven in Problem 1-(ii)&(iii)), it follows that

$$\Delta((A^\star, r^T A^\star), (A^\star, u^\star)) > \mu(n). \quad (6.5)$$

Lemma (HW3 Problem 1-(ii)&(iii)). Let $\mathcal{X}, \mathcal{Y}$ be distributions over a finite set and $\varepsilon \in (0, 1)$. Then

$$\Delta(\mathcal{X}, \mathcal{Y}) \leq \varepsilon \iff \forall t \in \mathbb{N}. \mathcal{X} \approx_{t, \varepsilon} \mathcal{Y}.$$

On the other hand, recall that Regev's PKE scheme (and $\text{LWE}_{n, m, q, \chi}$ as well) requires that $m = \Omega(n \log n)$. ($m \triangleq m_n$ is in fact a function of n .) Now we set $\varepsilon(n) \triangleq \frac{1}{2^n}$ and suppose that $m = \omega((n+1) \log q + 2n + 1)$. Then since $\mu(n) \geq \frac{1}{n^c} = \omega(\frac{1}{2^n})$ for infinitely many $n \in \mathbb{N}$, there must exist $n^* \in \mathbb{N}$ such that (1) $m \geq (n^* + 1) \log q + 2n^* + 1 = (n^* + 1) \log q + 2 \log(\frac{1}{\varepsilon(n^*)}) + 1$ and (2) $\mu(n^*) > \frac{1}{2^{n^*}}$ simultaneously. Finally, by Leftover Hash Lemma ([Theorem 2.1](#)), we have

$$\Delta((A^\star, r^T A^\star), (A^\star, u^\star)) \leq \varepsilon(n^*) = \frac{1}{2^{n^*}} < \mu(n^*). \quad (6.6)$$

Clearly (6.5) and (6.6) are contradictory, so we are done. ⊛

Reference

P1 (None)

P2 (None)

P3 (None)

P4 (None)

P5 (ii) [Lecture note by Vinod Vaikuntanathan](#)

P6 [Lecture note by Vinod Vaikuntanathan](#)